

# *Digital Medicine Authentication and Traceability Framework*



**ABSTRACT:** The rapid growth of the pharmaceutical supply chain has increased the risk of counterfeit, expired, and substandard medicines reaching consumers. Traditional batch-level tracking and manual verification methods are no longer sufficient to ensure medicine safety, transparency, and trust. This project proposes a Digital Medicine Authentication and Traceability Framework that provides end-to-end visibility of medicine movement from manufacturing to final consumption through the use of unique digital identifiers, and secure data management.

The framework assigns a unique digital identity to every medicine unit and records all lifecycle events, including manufacturing, distribution, retail transfer, and verification. A centralized verification server maintains the master records and validates all authenticity queries in real time. To improve efficiency in large-scale logistics, a parent-child packaging linkage model is used, where a single scan of a parent container automatically updates the status of all associated child units. This ensures full traceability while reducing operational workload.

The system supports a platform-independent verification interface, enabling stakeholders such as manufacturers, distributors, retailers, regulators, and consumers to authenticate medicines using web portals, enterprise systems, and optional mobile access. A secure request-response mechanism ensures that verification sessions are protected against replay, duplication, and unauthorized access. In the event of a defective or contaminated batch, the digital recall subsystem instantly flags all linked units, preventing further circulation.

The proposed framework enhances transparency, reduces counterfeit infiltration, and improves patient safety while remaining cost-effective and scalable. By combining traceability, and real-time verification into a unified digital ecosystem, this project offers a practical solution for building a safer and more trustworthy pharmaceutical supply chain.

# **I. Introduction and Literature Survey**

## **1. Global Problem and Need for Digital Traceability**

Counterfeit and substandard medicines are a major global health threat. According to the **World Health Organization (WHO)**, falsified medicines may contain wrong ingredients, incorrect dosages, or harmful substances and can lead to serious health consequences or death.

WHO Report:

<https://www.who.int/news-room/fact-sheets/detail/substandard-and-falsified-medical-products>

WHO further states that weak supply chain monitoring and lack of digital verification systems are key reasons behind the spread of counterfeit drugs, especially in developing countries.

## **2. International Standards and Regulations**

To reduce counterfeit medicine circulation, global standards and regulations have been introduced.

### **GS1 Healthcare Standards**

GS1 defines global standards for serialization and traceability using unique digital identifiers for medicines.

<https://www.gs1.org/industries/healthcare/traceability>

GS1 confirms that unit-level identification is the foundation of modern pharmaceutical traceability systems.

### **U.S. Drug Supply Chain Security Act (DSCSA)**

The FDA mandates complete drug traceability across the U.S. pharmaceutical supply chain.

<https://www.fda.gov/drugs/drug-supply-chain-integrity/drug-supply-chain-security-act-dscsa>

### **European Falsified Medicines Directive (FMD)**

The EU requires safety features and digital verification at the pharmacy level.

### **European Medicines Verification System (EMVS)**

A national-level verification system for medicines across Europe.

<https://emvo-medicines.eu>

This proves that digital medicine verification is already being used at national scale.

### **3. Existing Market Applications (Proof of Real Systems)**

- **Sproxil**

Consumer verification using unique codes.

<https://www.sproxil.com>

- **mPedigree (Goldkeys)**

SMS/USSD medicine verification used in Africa and Asia.

<https://mpedigree.com>

- **PharmaSecure – Verify My Medicine**

Online verification portal for consumers.

<https://verify.pharmasecure.com>

- **MedSnap**

AI-based pill image authentication.

<https://medsnap.com>

Although systems like **Sproxil**, **mPedigree**, **PharmaSecure**, **EMVS**, and **MediLedger** prove that medicine authentication is possible, they:

- Focus on only one part of the supply chain
- Do not offer parent-child packaging logic
- Are not open-source or platform-independent
- Do not provide real-time recall propagation
- Do not combine authentication and traceability into one framework

### **4. Blockchain & Digital Traceability Research**

#### **MediLedger Project**

Blockchain-based pharmaceutical traceability pilot.

<https://www.mediledger.com>

FDA Report on MediLedger:

<https://www.fda.gov/media/168283/download>

## 5. Academic & Government Research

### IEEE Secure Authentication Paper

“A Secure and Practical Authentication Scheme Using Personal Devices”

<https://ieeexplore.ieee.org/document/8269324>

This proves the importance of cryptographically secure verification.

### IEEE Pharma Supply Chain Security

<https://ieeexplore.ieee.org/document/9096841>

### Springer – Blockchain in Healthcare Supply Chain

<https://link.springer.com/book/10.1007/978-3-030-44007-3>

### India’s National Digital Health Mission

<https://ndhm.gov.in>

## 6. Identified Research Gap

- Most current platforms focus on **only one stage** of the pharmaceutical supply chain (consumer verification, regulatory compliance, or enterprise tracking), rather than providing **end-to-end traceability** from manufacturer to consumer.
- There is **no unified framework** that connects manufacturers, distributors, retailers, regulators, and consumers under a single verification and tracking platform.
- Current systems generally track medicines at **unit or batch level only** and do not implement a **parent-child packaging model** (box → strips → units) that enables bulk updates and efficient logistics.
- Many existing solutions are **closed, proprietary, and platform-dependent**, limiting interoperability, scalability, and affordability.
- Security and traceability are usually treated as **separate problems**; very few systems integrate **secure authentication mechanisms with lifecycle tracking** in a single operational framework.
- There is a lack of **open, scalable, and platform-independent frameworks** that can be deployed across diverse healthcare environments, including low-resource regions.

## 7. Motivation for Our Project

Our **Digital Medicine Authentication and Traceability Framework** is unique because:

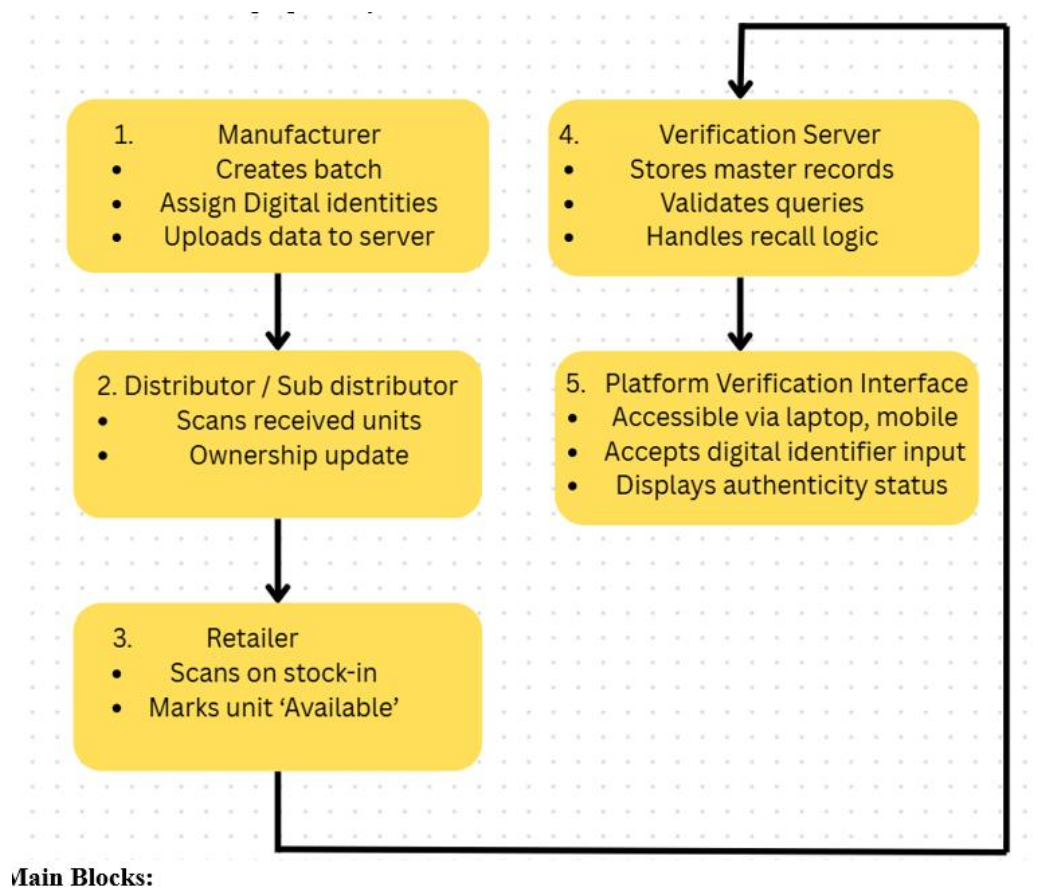
- It integrates manufacturer → distributor → retailer → consumer into one system
- It uses parent–child container logic
- It supports real-time digital recall
- It is platform-independent
- It is open and scalable

This directly addresses the limitations found in existing systems and research.

## II. Functional Flow Chart

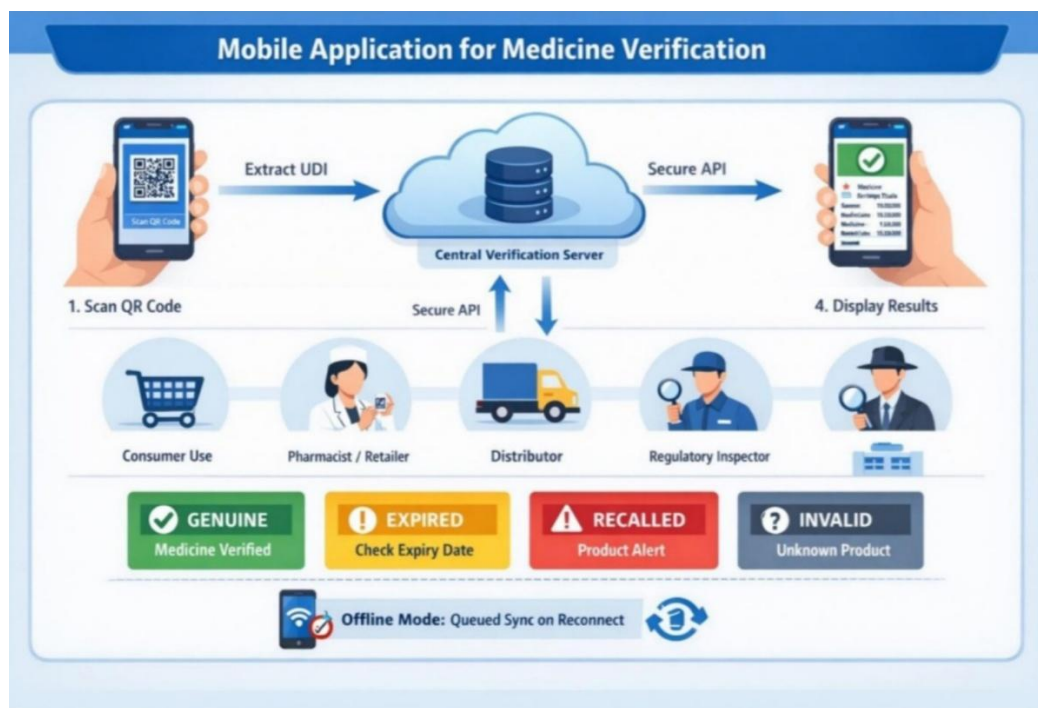
The Digital Medicine Authentication and Traceability Framework is designed to create a transparent and secure environment for tracking medicines throughout the pharmaceutical supply chain. It ensures that every medicine unit can be uniquely identified, verified, and monitored from the point of production to the point of sale. The system connects all stakeholders—manufacturers, distributors, retailers, regulators, and consumers.

The diagram shows the complete working flow of the **Digital Medicine Authentication and Traceability Framework**.



- The process starts with the Manufacturer, who creates a batch, assigns unique digital identities to each medicine unit, and uploads all details to the central server.
- The Distributor / Sub-distributor receives the medicines, scans the units, and updates ownership and movement information in the system.
- The Retailer scans medicines during stock-in and marks them as available for sale, acting as the final authorized checkpoint before consumer access.

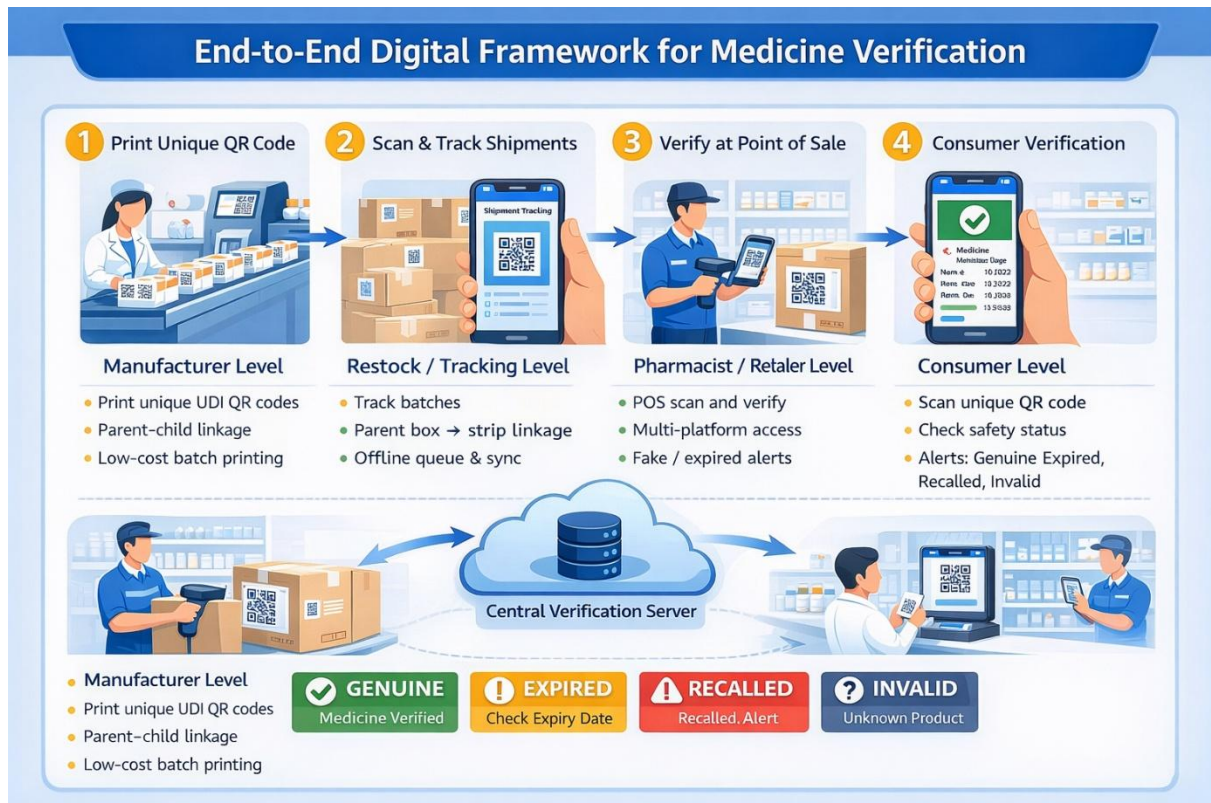
- The Verification Server stores all master records, validates authenticity queries, and manages recall logic when unsafe batches are identified.
- The Platform Verification Interface allows users to access the system through mobile devices, accept digital identifiers, and display authenticity status.
- Each stakeholder interacts with the verification server to ensure real-time tracking and transparency.
- The entire framework ensures that medicine movement is digitally recorded from manufacturing to retail, providing full traceability and safety.



**This figure shows** the working of a mobile application for medicine verification. The user scans the QR code on the medicine pack, and the app extracts the unique identification (UDI) and sends it securely to a central verification server through an API. The server checks the details and sends the result back to the app, which is then displayed on the screen.

The system can be used by **consumers, pharmacists, distributors, and regulatory inspectors**. Based on the verification, the medicine status is shown as **Genuine, Expired, Recalled, or Invalid**. The app also supports **offline mode**, where scans are stored and synchronized when the internet connection is restored.





This diagram shows the **End-to-End Digital Framework for Medicine Verification** used in the MedTrace system. It explains how a medicine is tracked and verified from the **manufacturer to the final consumer** using unique QR codes and a central server.

At the **Manufacturer Level**, each medicine pack is printed with a **unique UDI QR code**. The system also creates a **parent-child hierarchy**, linking cartons, boxes, and strips together. This makes every unit traceable at a very low cost.

At the **Restock and Tracking Level**, distributors and warehouses **scan shipments** to update stock movement. Each scan connects the parent box to its child strips and sends data to the **central verification server**, even supporting offline sync.

At the **Pharmacist / Retailer Level**, medicines are scanned at the **POS counter** before sale. The system instantly checks if the product is **genuine, expired, recalled, or invalid**, helping prevent fake or unsafe medicines from being sold.

At the **Consumer Level**, customers can scan the QR code using the mobile app to verify authenticity. They receive a real-time safety status such as:

- **Genuine** – medicine is verified
  - **Expired** – check expiry date
  - **Recalled** – do not consume
  - **Invalid** – unknown or fake product.
- All scans connect to a **Central Verification Server**, which acts as the single source of truth.



### III. Prototype/Website of our Project

This figure shows the desktop **landing page of the proposed Digital Medicine Authentication and Traceability Framework web platform**. It is the main entry point for all stakeholders to access the system and its services.

From this page, authorized users such as manufacturers, distributors, retailers, and regulators can **log in using the “Partner Login” option** to access their respective dashboards. New organizations can also request access or view system documentation through the provided buttons.

MedTrace

Solutions
Network
Pricing

Partner Login

SUPPLY CHAIN INTEGRITY ACTIVE

# Secure Your Pharma Supply Chain

End-to-end blockchain traceability for Manufacturers, Distributors, and Retailers. Eliminate counterfeits and ensure compliance with FDA & WHO standards.

Request Demo
View Documentation

TRUSTED BY INDUSTRY LEADERS

Pfizer
Novartis
Cipla
Sun Pharma

PROCESS

## End-to-End Traceability

1

### Manufacturing

Unique QR generation at the source.

2

### Distribution

IoT monitoring during transit.

3

### Pharmacy

Verification upon receipt.

4

### Consumer

Instant scan verification via app.

## Enterprise Solutions

Scalable tools for the modern pharma ecosystem.

### Connected Global Network

Real-time data synchronization across borders ensuring medicine safety everywhere.

### Logistics Tracking

Real-time GPS and condition monitoring for cold-chain and sensitive pharmaceutical shipments globally.

### Blockchain Ledger

Immutable record keeping for every single unit from factory to pharmacy. Fraud-proof architecture.

### Sales Analytics

Deep AI-driven insights into consumption patterns and counterfeit attempt heatmaps by region.

99.9%  
UPTIME

50M+  
UNITS TRACKED

20+  
COUNTRIES

0%  
COUNTERFEIT RATE

### Global Compliance

Built to meet the rigorous standards of global health authorities.

FDA 21 CFR
EU FMD
DSCSA

### Seamless Integrations

Works with your existing ERP and warehouse management systems.

SAP
Oracle
Microsoft

## Trusted by Pharma Executives

"Before integrating MedTrace, we struggled with significant blind spots in our supply chain, especially during last-mile delivery. Now, we have granular visibility that allows us to track every single step with 100% accuracy. It has completely transformed our operations efficiency."

**Rohit Sharma**  
DIRECTOR SALES, Cipla

"The threat of counterfeit drugs entering our supply chain was keeping me up at night. Since deploying MedTrace's blockchain ledger, we have an immutable record of every product journey. We have seen a complete elimination of verification disputes in just six months."

**Dr. Elvira B.**  
DIRECTOR HEAD, NOVARTIS

"What impressed me most was the speed of integration. We connected MedTrace with our complex SAP ERP environment in less than two weeks without any downtime. The real-time analytics dashboard gave us proactive insights that we never had before."

**Vikram M.**  
S. CHAKRA, SUN PHARMA

"Scalability was our biggest concern, but MedTrace proved its robustness immediately. We expanded our tracking to over 30 countries in a single quarter. The API is incredibly well-documented, and the support team is proactive and knowledgeable."

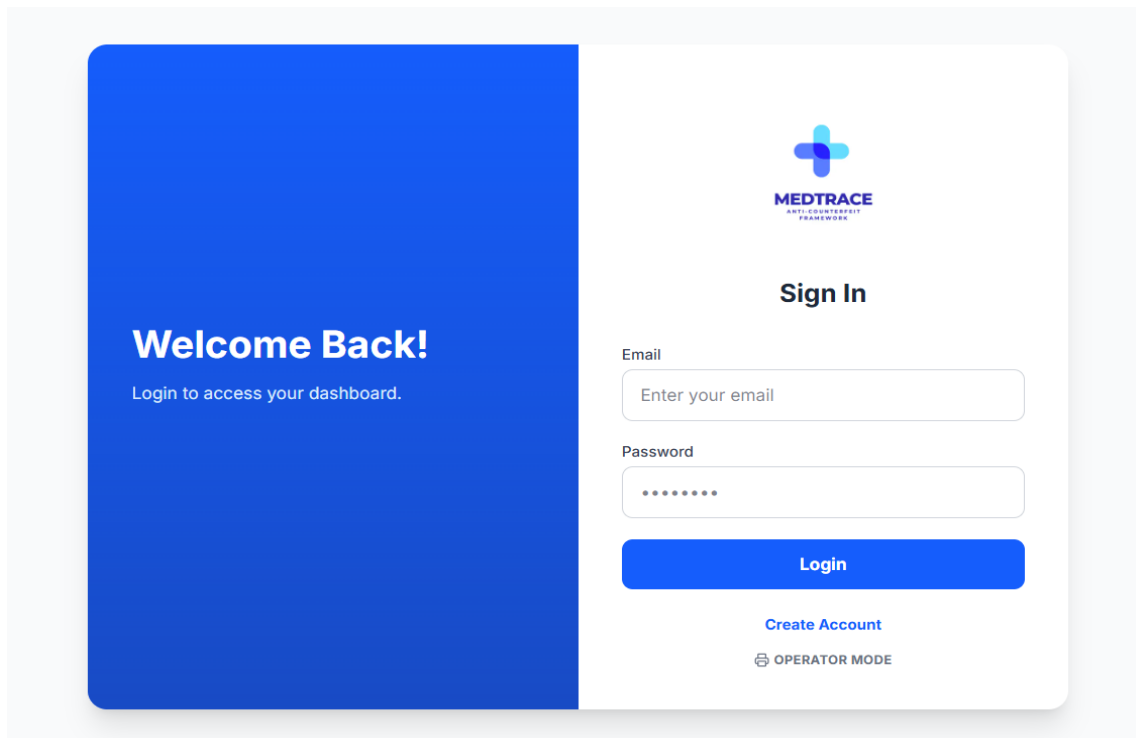
**Sarah Johnson**  
CTO, GLOBAL HEALTH

"For cross-border pharmaceutical shipments, compliance is everything. MedTrace automated our entire regulatory reporting process for FDA and EU FMD standards. It saves us thousands of man-hours every year and ensures we are always audit-ready."

**Ajay and Khushi**  
DIRECTORS, ZENITH

"The core and backbone of our business is the quality of our products. MedTrace ensures the quality of our products is maintained throughout the entire supply chain, from manufacturing to the patient's hands."

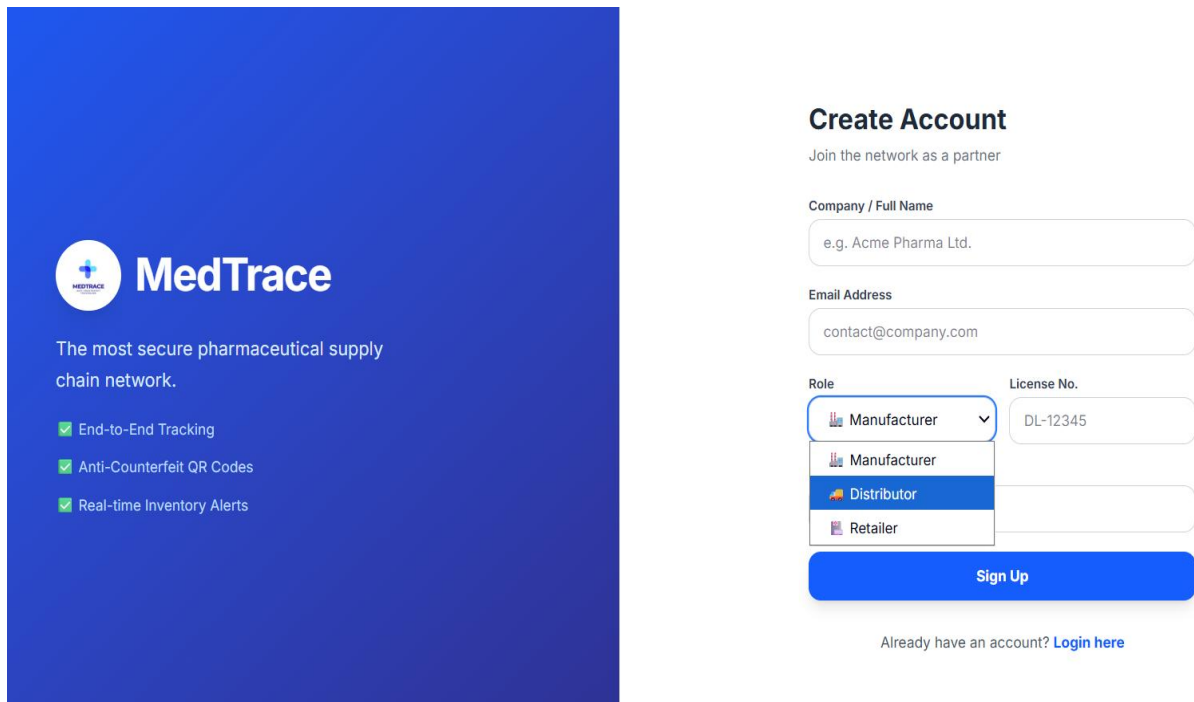
**Vishal K.**  
DIRECTOR, SUN PHARMA



This figure shows the **secure login interface** of the Digital Medicine Authentication and Traceability Framework. It is the authentication gateway that allows authorized users to access the system dashboard.

Users such as manufacturers, distributors, retailers, and regulatory operators can enter their **registered email and password** to log in. After successful authentication, the system grants role-based access to ensure that each user can only perform actions permitted for their role.

The interface also provides options to **create a new account** for onboarding new organizations and to enter **operator mode** for field-level verification tasks. The clean and simple layout ensures ease of use while maintaining data security.



The image displays the MedTrace registration interface. On the left, a blue banner features the MedTrace logo and the text "The most secure pharmaceutical supply chain network." Below this, three green checkmarks list features: "End-to-End Tracking", "Anti-Counterfeit QR Codes", and "Real-time Inventory Alerts". On the right, a white form titled "Create Account" with the subtitle "Join the network as a partner" contains the following fields: "Company / Full Name" (with placeholder "e.g. Acme Pharma Ltd."), "Email Address" (with placeholder "contact@company.com"), a "Role" dropdown menu (with options: Manufacturer, Manufacturer, Distributor, and Retailer; the first "Manufacturer" option is selected), and "License No." (with placeholder "DL-12345"). A blue "Sign Up" button is at the bottom of the form, and a link "Already have an account? Login here" is below it.

This figure shows the **registration interface** of the Digital Medicine Authentication and Traceability Framework. It allows new organizations to join the system as authorized partners.

Users are required to enter their **company or full name**, **email address**, select their **role** (Manufacturer, Distributor, or Retailer), and provide a valid **license number**. This ensures that only verified and legitimate stakeholders are allowed to participate in the network.

## 1. Manufacturer Login Page

This figure shows the manufacturer dashboard of the Digital Medicine Authentication and Traceability Framework. It is the main control panel for a manufacturer after logging into the system.

MedTrace

MANUFACTURER

Trusted Manufacturer

Dashboard

Track Medicine

PRODUCTION

Product Catalog

Production & Print

Create Batch

My Inventory

Shipments

Sales Orders

ROLE MANUFACTURER

Sign Out

APOLLO

VERIFIED MANUFACTURER

ADDRESS NOT UPDATED

EDIT PROFILE

DRUG LICENSE

WB-2348

GST NUMBER

NOT SET

CONTACT

NOT SET

EMAIL

bose001@gmail.com

SYSTEM OPERATIONAL

Overview

Welcome back, Apollo

31 Jan 2026

View Reports

TOTAL REVENUE

₹8,957

+12.5%

TOTAL UNITS

95

In Inventory

DISTRIBUTORS

2

Active Partners

SYSTEM ALERTS

0

All Clear

Revenue Trend

MONTHLY PERFORMANCE (JAN - DEC)

Revenue Trend (Monthly)

₹10k

₹7.5k

₹5k

₹2.5k

₹0k

Jan

Feb

Mar

Apr

May

Jun

Jul

Aug

Sep

Oct

Nov

Dec

Inventory Mix

STOCK BY MEDICINE TYPE

Stock Distribution

3 CATEGORIES

TABLET

SYRUP

INJECTION

Weekly Performance

LAST 7 DAYS SALES

10000

7500

5000

2500

0

Sun

Mon

Tue

Wed

Thu

Fri

Sat

Top Products

View All

#1

Diphenhydramine

SYRUP • 40 SOLD

₹5,200

#2

Calpol

TABLET • 20 SOLD

₹1,600

#3

Neomol Inj

INJECTION • 5 SOLD

₹1,150

#4

Dolo

TABLET • 10 SOLD

₹800

Production vs Sales Analysis

LAST 6 MONTHS VOLUME COMPARISON

180

135

90

45

0

Aug

Oct

Dec

Jan

Production

Sales

Action Needed

LOW STOCK

Dolo

70 Left

LOW STOCK

Diphenhydramine

0 Left

LOW STOCK

Calpol

20 Left

EXPIRING

Dolo

2/2/2026

Top Partners

S

Saha

₹6,357 Val

#1

R

Roy

₹2,600 Val

#2

Recent Activity

Order Dispatched

To Distributor • ₹1,357

06:18 AM

Order Dispatched

To Distributor • ₹1,600

06:17 AM

Order Dispatched

To Distributor • ₹2,600

06:12 AM

Order Dispatched

To Distributor • ₹2,600

06:08 AM

Order Dispatched

To Distributor • ₹800

06:04 AM

The dashboard provides an overview of the manufacturer's profile and system status, including verification status, license details, and contact information. It allows the manufacturer to manage production, track medicine batches, monitor inventory, and view distributor and shipment activity.

The left navigation menu helps the user access important functions such as creating batches, printing digital identities, tracking medicines, managing shipments, and checking system alerts.

The dashboard also displays graphical summaries and alerts that help the manufacturer quickly understand the current system condition and take necessary actions. Overall, this page acts as a centralized workspace for managing all manufacturing and traceability operations within the platform.

The dashboard also highlights important alerts such as low stock and expiring products, enabling timely actions to prevent supply disruptions. Recent activity updates help the manufacturer track shipment movements and system interactions. Overall, this dashboard improves operational efficiency by providing a centralized and easy-to-use workspace for managing production, tracking medicines, and ensuring supply chain transparency.

**+ Add New Medicine**  
ENTER PRODUCT DETAILS BELOW

|                                |                        |                              |
|--------------------------------|------------------------|------------------------------|
| PRODUCT CODE<br>Auto Generated | BRAND NAME<br>Pantocid | GENERIC NAME<br>Pantoprazole |
|--------------------------------|------------------------|------------------------------|

MEDICINE TYPE  
Injection

**PACKAGING LOGIC:** ☐ Single Unit (1 Vial/Box) ☒ Multi Pack (Many Ampules/Box)

|                                                               |                  |                                                                                |
|---------------------------------------------------------------|------------------|--------------------------------------------------------------------------------|
| AMPULES PER BOX<br>5<br><small>(vials count)</small><br>UNITS | STRENGTH<br>40mg | BASE PRICE (DISTRIBUTOR)<br>₹ 120<br><small>Cost + Manufacturer Profit</small> |
|---------------------------------------------------------------|------------------|--------------------------------------------------------------------------------|

STORAGE TEMP  
20

**+ Add to Catalog**

This screen is used to add a new **Injection medicine** with real-world packaging logic.

- We are adding an injection medicine **Pantocid (Pantoprazole)**.
- The **Medicine Type** is selected as **Injection**.



- The **Packaging Logic** is set to **Multi Pack (Many Ampoules/Box)**, which means one box contains multiple ampoules.
  - **Ampoules per box = 5**, so each box has 5 injection vials.
  - **Strength = 40 mg**, which is the dose of each ampoule.
  - **Base Price (Distributor) = ₹120**, which is the price of one box.
  - **Storage Temperature = 20°C**, showing the required storage condition.
- This setup follows the same way injections are packed in real life. It allows the system to digitally track medicines at **box and ampoule level**, making the supply chain more accurate and traceable.

The screenshot shows a web form titled '+ Add New Medicine' with the subtitle 'ENTER PRODUCT DETAILS BELOW'. The form contains several input fields and a button:

- PRODUCT CODE:** A text field with the placeholder 'Auto Generated'.
- BRAND NAME:** A text field containing 'Pantocid'.
- GENERIC NAME:** A text field containing 'Pantoprazole'.
- MEDICINE TYPE:** A dropdown menu with 'Injection' selected.
- PACKAGING LOGIC:** A section with two radio buttons: 'Single Unit (1 Vial/Box)' (which is selected) and 'Multi Pack (Many Ampoules/Box)'.
- STRENGTH:** A text field containing '40mg'.
- BASE PRICE (DISTRIBUTOR):** A text field containing '₹ 120'. Below this field is the text 'Cost + Manufacturer Profit'.
- STORAGE TEMP:** A text field containing '20'.
- Button:** A large blue button at the bottom with the text '+ Add to Catalog'.

This page shows that we add a new **Injection medicine** with **single-unit packaging**.

- The medicine being added is **Pantocid (Pantoprazole)**.
- The **Medicine Type** is selected as **Injection**.
- The **Packaging Logic** is set to **Single Unit (1 vial per box)**, which means each box contains only one injection vial.
- **Strength = 40 mg**, showing the dosage of the injection.
- **Base Price (Distributor) = ₹120**, which is the price of one box containing one vial.
- **Storage Temperature = 20°C**, indicating the required storage condition.

This option is used for injections that are sold **one vial per box**, and it helps the system track each unit accurately from manufacturer to retailer.

+ Add New Medicine  
ENTER PRODUCT DETAILS BELOW

PRODUCT CODE

Auto Generated

BRAND NAME

Allerzin

GENERIC NAME

Cetirizine

MEDICINE TYPE

Syrup

STRENGTH

50mg

BASE PRICE (DISTRIBUTOR)

₹ 130

STORAGE TEMP

25

+ Add to Catalog

This page shows that we can add a new **syrup medicine** to the system.

- The **Brand Name** is **Allerzin** and the **Generic Name** is **Cetirizine**.
- The **Medicine Type** is selected as **Syrup**, which means the medicine is in liquid form and sold in bottles.
- The **Strength** is **50 mg**, indicating the dosage of the syrup.
- The **Base Price (Distributor)** is **₹130**, which is the cost of one bottle.
- The **Storage Temperature** is set to **25°C**, showing normal room-temperature storage.

After clicking **Add to Catalog**, this syrup will be registered in the system so that QR codes, batch tracking, and supply-chain verification can be applied to this product.

+ Add New Medicine  
ENTER PRODUCT DETAILS BELOW

PRODUCT CODE

Auto Generated

BRAND NAME

Dolo

GENERIC NAME

Paracetamol

MEDICINE TYPE

Tablet

STRIP SIZE

10

(tablets per strip)

STRENGTH

650mg

BASE PRICE (DISTRIBUTOR)

₹ 85

STORAGE TEMP

25

+ Add to Catalog

This page is used to add a new **tablet medicine** to the system.

- The **Brand Name** is **Dolo** and the **Generic Name** is **Paracetamol**.
- The **Medicine Type** is selected as **Tablet**, which means the medicine is packed in strips.

- The **Strip Size** is **10 tablets per strip**, matching standard tablet packaging.
- The **Strength** is **650 mg**, which shows the dosage of each tablet.
- The **Base Price (Distributor)** is **₹85**, which is the cost of one strip.
- The **Storage Temperature** is set to **25°C**, indicating normal room-temperature storage.

After clicking **Add to Catalog**, this tablet will be registered in the system and will follow the same QR-based tracking, batch linking, and verification process across the supply chain.

**+ Add New Medicine**  
ENTER PRODUCT DETAILS BELOW

|                                |                      |                                   |
|--------------------------------|----------------------|-----------------------------------|
| PRODUCT CODE<br>Auto Generated | BRAND NAME<br>Volini | GENERIC NAME<br>Diclofenac Sodium |
| MEDICINE TYPE<br>Spray         | STRENGTH<br>75 mg    | BASE PRICE (DISTRIBUTOR)<br>₹ 60  |
| STORAGE TEMP<br>25             |                      |                                   |

**+ Add to Catalog**

This page is used to add a new **spray-type medicine** to the system.

- The **Brand Name** is **Volini** and the **Generic Name** is **Diclofenac Sodium**.
- The **Medicine Type** is selected as **Spray**, which means the product is packed in a spray bottle for topical use.
- The **Strength** is **75 mg**, indicating the active drug content.
- The **Base Price (Distributor)** is **₹60**, which is the cost of one spray unit.
- The **Storage Temperature** is set to **25°C**, showing it can be stored at normal room temperature.

After clicking **Add to Catalog**, this spray will be registered in the system and will be tracked with unique QR codes, batch details, and verification across the supply chain.

Product Catalog

Manage your medicine master catalog and set base prices.

Grid

List

Add New Medicine

ENTER PRODUCT DETAILS BELOW

PRODUCT CODE

Auto Generated

BRAND NAME

Cipolex

GENERIC NAME

Eye drop

MEDICINE TYPE

Drops

STRENGTH

0.3%

BASE PRICE (DISTRIBUTOR)

₹ 15

Cost + Manufacturer Profit

STORAGE TEMP

20

+ Add to Catalog

This page is used to add a new **eye drop medicine** to the product catalog.

- The **Brand Name** is **Cipolex** and the **Generic Name** is **Eye drop**.
- The **Medicine Type** is selected as **Drops**, which means the product is packed in a liquid dropper bottle.
- The **Strength** is **0.3%**, indicating the concentration of the active ingredient.
- The **Base Price (Distributor)** is **₹15**, which is the cost of one bottle at distributor level.
- The **Storage Temperature** is set to **20°C**, meaning the drops should be stored in a cool, room-temperature environment.

After clicking **Add to Catalog**, this eye-drop product will be saved in the system and can be tracked with batch details, QR codes, and verification across the supply chain.

**MedTrace**  
MANUFACTURER

Treated Manufacturer

Track Medicine

PRODUCTION

Product Catalog

Production & Print

Create Batch

My Inventory

Shipments

Sales Orders

Quality & Recall

ROLE

MANUFACTURER

Sign Out

**APOLLO**
VERIFIED MANUFACTURER

ADDRESS NOT UPDATED

EDIT PROFILE

DRUG LICENSE  
**WB-2346**

GST NUMBER  
**NOT SET**

CONTACT  
**NOT SET**

EMAIL  
**bose001@gmail.com**

**Product Catalog**
Manage your medicine master catalog and set base prices.

Grid
List

+ Add New Medicine
ENTER PRODUCT DETAILS BELOW

PRODUCT CODE
Auto Generated

BRAND NAME
Cipolex

GENERIC NAME
Eye drop

MEDICINE TYPE
Drops

STRENGTH
0.3%

BASE PRICE (DISTRIBUTOR)
₹ 15

STORAGE TEMP
20

+ Add to Catalog

**Volini**  
Diclofenac Sodium

TYPE
SPRAY

STRENGTH
75 mg

BASE PRICE
₹60.00

**Zelocip CF Cream**  
Flucinolone

TYPE
CREAM

STRENGTH
25 g

BASE PRICE
₹35.00

**Cipolex**  
Eye drops

TYPE
DROPS

STRENGTH
0.3%

BASE PRICE
₹15.00

**Dolo**  
Paracetamol

TYPE
TABLET

PACK SIZE
5 / Strip

STRENGTH
650 mg

BASE PRICE
₹80.00

**Neomol Inj**  
Paracetamol

TYPE
INJECTION

PACK SIZE
10 / Box

STRENGTH
1g/100 ml

BASE PRICE
₹210.00

**Diphenhydramine**  
Benadryl

TYPE
SYRUP

PACK SIZE
10 / Strip

STRENGTH
150 mg

BASE PRICE
₹130.00

**Calpol**  
Paracetamol

TYPE
TABLET

PACK SIZE
10 / Strip

STRENGTH
650 mg

BASE PRICE
₹80.00

This figure shows the total Product Catalog page of the manufacturer module in the Digital Medicine Authentication and Traceability Framework. It is used by the manufacturer to manage the master list of medicines before production and batch creation.

From this page, the manufacturer can add new medicines by entering details such as brand name, generic name, medicine type, strip size, strength, storage temperature, and base price. Once added, each product becomes part of the system's master catalog.

The lower section displays all existing products in a card format, showing their basic information such as medicine type, pack size, strength, and base price. This allows the manufacturer to easily view, update, and

manage all registered medicines. All the medicine details that are added previously are shown here.

This page ensures that only verified and standardized product information is used when creating batches and assigning digital identities, supporting accurate traceability throughout the supply chain.

The screenshot displays the MedTrace MANUFACTURER interface. The sidebar on the left contains navigation links: Track Medicine, Product Catalog, Production & Print (highlighted), Create Batch, My Inventory, Shipments, Sales Orders, and Quality & Recall. The main content area is titled 'Team Management' with the subtitle 'Manage access for your production floor staff.' It includes a 'Register New Operator' form with fields for 'Full Name' (e.g., Rahul Das) and 'Access PIN' (4-6 digits). A blue button 'Create Operator Account' is at the bottom of the form. To the right, a table titled 'Search operators by name...' shows 'No operators found' with a message to 'Add your first team member using the form.'

This page is used to manage production floor operators and assign controlled access for printing digital identities on cartons and strips. From this interface, the manufacturer can create a secure operator account by entering the operator's name and generating a unique access PIN. The assigned operator can then log in using this PIN to perform printing tasks. The system allows the manufacturer to assign specific production jobs and define a limited time window for completing the printing process. Once the time limit expires or the assigned task is completed, the operator's access is automatically disabled. This ensures that printing privileges cannot be misused and that every printed digital identity is properly authorized and traceable.

By controlling operator access in this manner, the framework strengthens security on the production floor.

**Team Management**  
Manage access for your production floor staff.

**Register New Operator**  
Create a secure access account for machinery.

Full Name  
Suvik

Access PIN 4-6 DIGITS  
1 2 3 4 5

The operator will use this PIN to log in.  
⚠ Make sure to note it down.

**Create Operator Account**

Search operators by name...

| OPERATOR PROFILE                                                                | STATUS | DATE ADDED | ACTIONS |
|---------------------------------------------------------------------------------|--------|------------|---------|
| <p><b>No operators found</b><br/>Add your first team member using the form.</p> |        |            |         |

**Assign Production** →

**TOTAL STAFF**  
0

ROLE  
MANUFACTURER

This page is the **Team Management** section for the manufacturer. It is used to manage access for production floor staff who operate machines and handle printing or batch work.

On the left, the **Register New Operator** form allows the admin to create a secure operator account. The admin enters the **full name** of the operator and sets a **4–6 digit access PIN**. This PIN is used by the operator to log in to the system or machine. A warning reminds the admin to note the PIN, as it is required for access.

After clicking **Create Operator Account**, the operator will be added to the system.

On the right side, there is a **staff list panel** that shows all registered operators, along with their **status, date added, and actions**. Currently, no operators are listed.

At the top, the **Assign Production** button is used to assign operators to specific production or printing tasks. The **Total Staff** counter shows how many operators are registered.

The screenshot shows the MedTrace APOLLO interface for a manufacturer. The top header includes the MedTrace logo, the manufacturer name 'APOLLO', and a status 'VERIFIED MANUFACTURER'. Below this, there's a section for 'ADDRESS NOT UPDATED' and fields for 'DRUG LICENSE' (WB-2346), 'GST NUMBER' (NOT SET), 'CONTACT' (NOT SET), and 'EMAIL' (bose001@gmail.com). An 'EDIT PROFILE' button is in the top right. The left sidebar contains navigation links: Dashboard, Track Medicine, Product Catalog, Production & Print (highlighted), Create Batch, My Inventory, Shipments, and Sales Orders. The main content area is titled 'Production Control' and features two panels. The 'Assign New Job' panel on the left allows creating a task for the production floor with fields for 'ASSIGN TO (OPERATOR)' (Souvik), 'SELECT BATCH' (B-202601-004-D1C5 - Dolo (80)), 'TARGET QTY' (80), 'MACHINE / LINE' (Line 2 (Standard)), and 'ACCESS VALIDITY' (8 hours, with a note 'Expires in 8 Hours'). A 'Generate Job Code' button is at the bottom. The 'Live Operators' panel on the right shows 'No active jobs. Assign a job to see it here.'

This page is the **Production Control** screen that appears after clicking **Assign Production** from Team Management.

From this page, the manufacturer can **assign a live production job** to a specific operator and machine line.

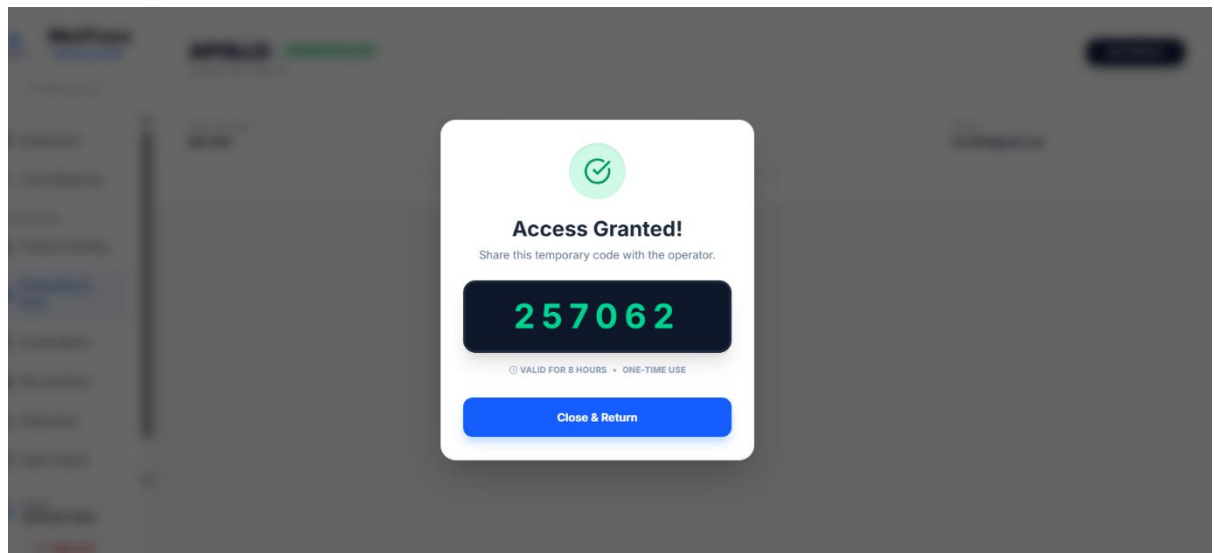
On the left side, the **Assign New Job** panel is used to create a task for the production floor:

- **Assign to Operator:** Selects which operator will handle the job.
- **Select Batch:** Chooses the medicine batch that needs QR printing or packaging.
- **Target Quantity:** Defines how many units need to be produced or labeled.
- **Machine / Line:** Selects the production or printing line where the job will run.
- **Access Validity:** Sets how long the operator can access this job (for example, 8 hours). This improves security by limiting time-based access.

After filling these details, the admin clicks **Generate Job Code**. This code is then used by the operator to start the assigned task on the production floor.

On the right side, the **Live Operators** panel shows all currently active jobs. It will display running tasks once operators begin work.

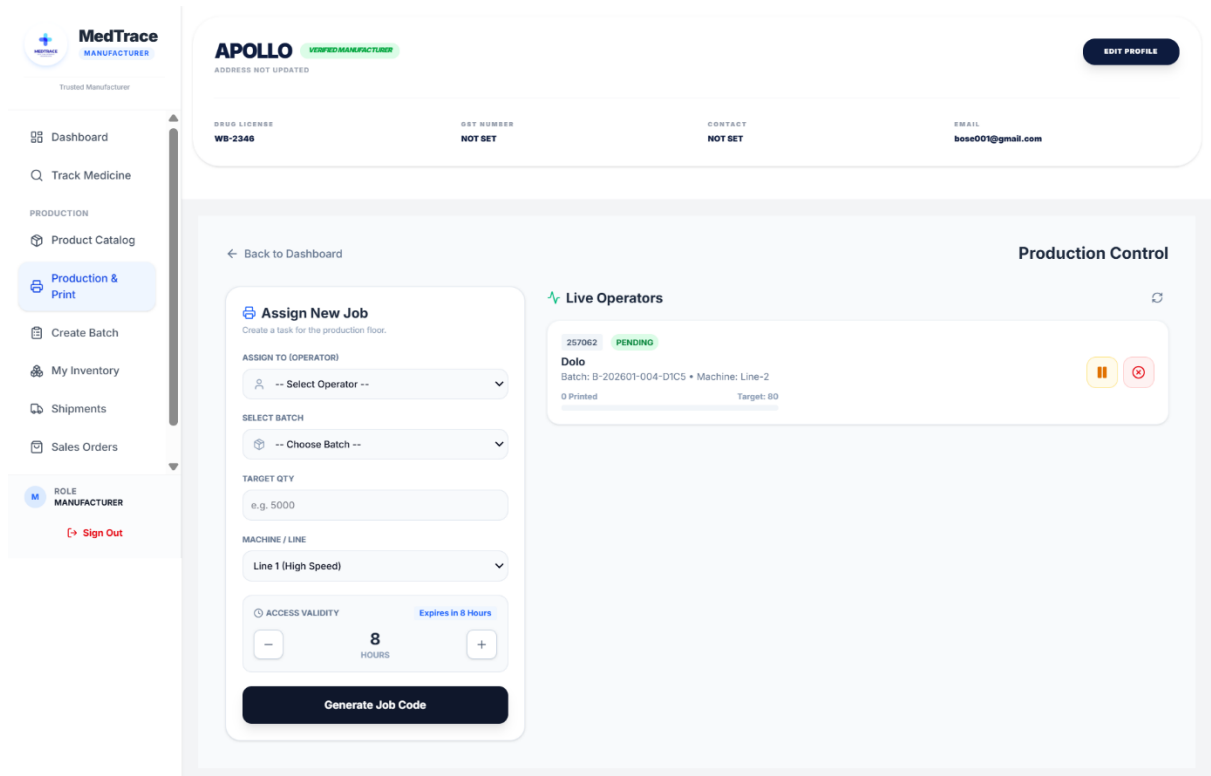




This popup appears **after the manufacturer generates a job code** from the Production Control page.

It confirms that **temporary access has been successfully created** for the operator.

- The system generates a **one-time job access code** (e.g., 257062).
- This code must be **shared with the assigned operator**.
- The code is **valid only for 8 hours** and can be used **once**, which prevents misuse.
- The operator enters this code on the production terminal to start the assigned job.



After the **job code is generated and shared with the operator**, the task appears in the **Live Operators** panel on the Production Control page.

This screen shows the **real-time production job status**.

**What this means:**

- The job is now in **PENDING** state, which means the operator has not started printing yet.
- The system displays:
  - **Job Code:** 257062
  - **Medicine:** Dolo
  - **Batch:** B-202601-004-D1C5
  - **Machine:** Line-2
  - **Target Quantity:** 80 units
  - **Printed:** 0 (progress starts from here)

**Controls on the right:**

- **|| Pause** – temporarily stops the job.
- **✗ Cancel** – stops and invalidates the job.

As soon as the operator logs in using the job code and starts printing, the **printed count and progress bar will update in real time**.

The screenshot displays the MedTrace MANUFACTURER interface. On the left is a sidebar with navigation options: Dashboard, Track Medicine, Product Catalog, Production & Print (highlighted), Create Batch, My Inventory, Shipments, and Sales Orders. Below these is a 'ROLE MANUFACTURER' section with a 'Sign Out' button. The main content area is titled 'Team Management' and includes a sub-header 'Manage access for your production floor staff.' and an 'Assign Production' button. A 'Register New Operator' form is visible, with fields for 'Full Name' (e.g., Rahul Das) and 'Access PIN' (4-6 digits). Below the form is a 'Create Operator Account' button. To the right of the form is a table of registered operators.

| OPERATOR PROFILE        | STATUS | DATE ADDED   | ACTIONS |
|-------------------------|--------|--------------|---------|
| Souvik<br>op-12345-b091 | ACTIVE | Jan 31, 2026 |         |

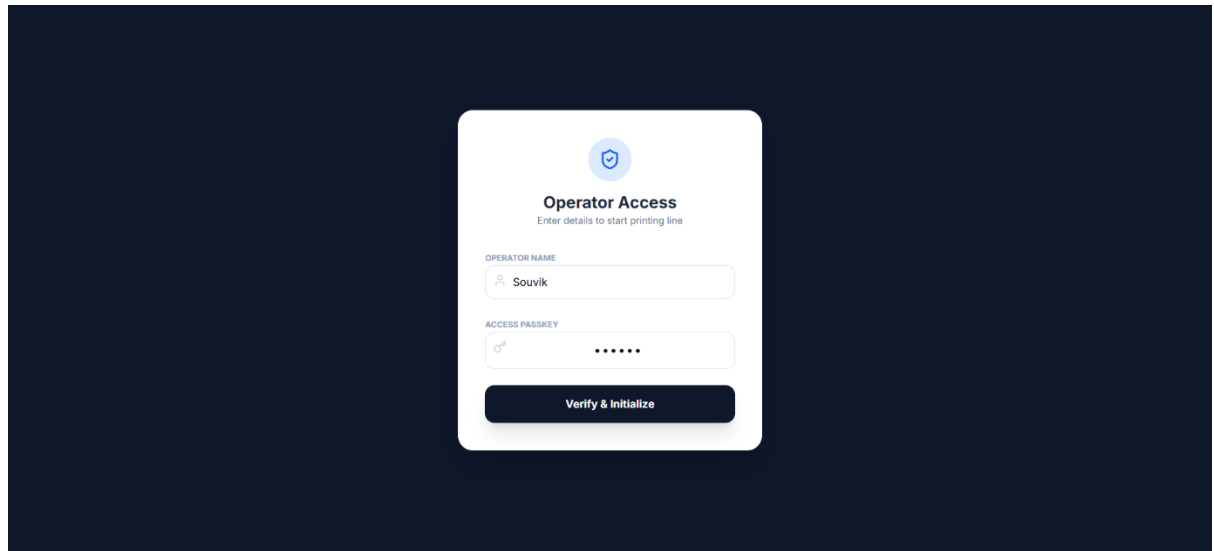
Here we can see the registered operator:

- **Name:** Souvik
- **Operator ID:** op-12345-b091
- **Status:** Active
- **Date Added:** Jan 31, 2026

This confirms that the operator is now **authorized to work on the production line**.

Only operators added here can:

- Receive **job codes**
- Access **printing machines**
- Print **QR codes linked to batches**



This screen is the **secure operator authentication page** used on the production floor before starting the QR code printing process.

The operator must log in using:

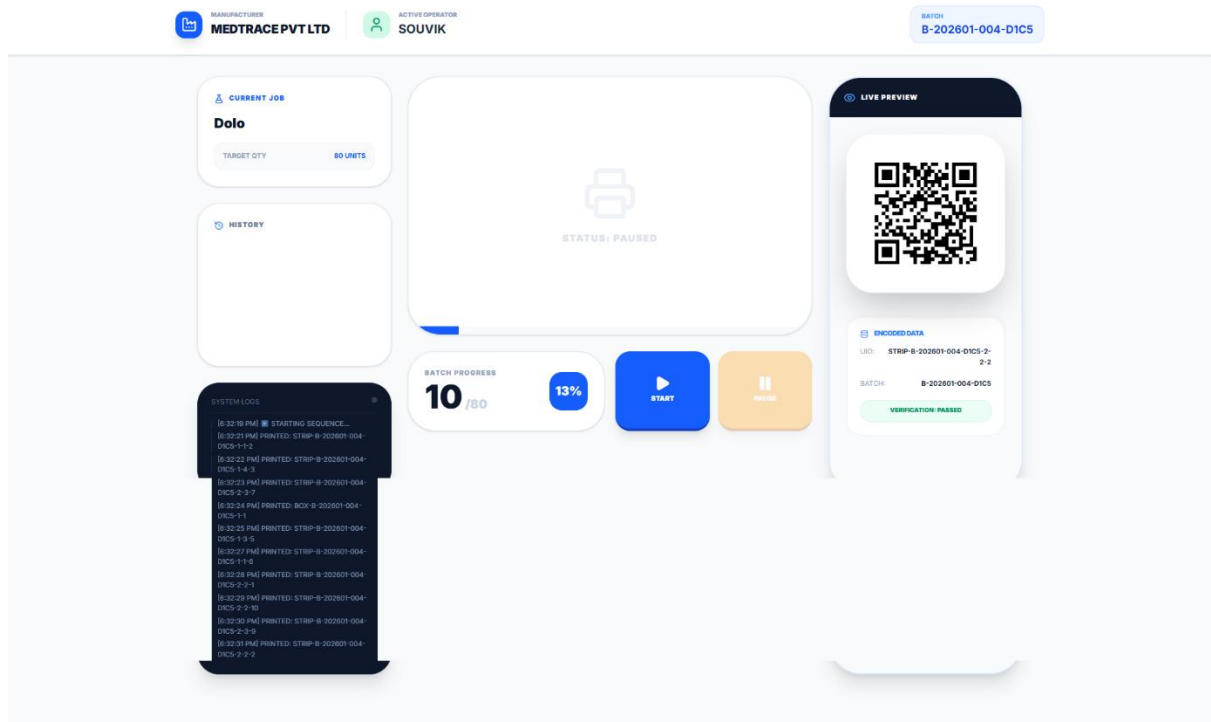
- **Operator Name** – selected or entered by the manufacturer (e.g., *Souvik*)
- **Access Passkey (PIN)** – the secret 4–6 digit code created in the Team Management page

After entering these details, the operator clicks **“Verify & Initialize.”**

Once verified:

- The operator is connected to the assigned **job code**
- The printing line is **unlocked**
- The system starts **real-time tracking** of how many QR codes are printed

This step completes the security chain between **manufacturer → operator → machine → batch → QR code**.



This screen is the **live production control panel** used by the operator during QR code printing. It shows real-time job status, QR generation, and batch progress.

#### **Live Preview (Right Panel)**

- Displays the **exact QR code** being printed.
- Shows **encoded data** like:
  - Unique UID (strip-level code)
  - Batch number
- **Verification Passed** confirms the QR is valid before printing.

#### **Central Status Area**

- Shows machine state (Paused / Running).
- Operator can **Start** or **Pause** the printing process.

### Job Completion & Auto Removal (Manufacturer View)

After the operator finishes printing the **target quantity**, the system **automatically closes the job**.

On the **Production Control** page:

- The job is **removed from Live Operators**.
  - The screen shows **“No active jobs”**.
  - This confirms that the batch printing is **fully completed**.
  - No further QR codes can be printed under that job.
  - The job record is safely stored in the system logs for audit and tracking.
- This ensures that **each batch is printed only once**, prevents over-printing, and keeps the production line **secure, controlled, and traceable**.

**MedTrace**  
MANUFACTURER

**APOLLO**  
ADDRESS NOT UPDATED

DRUG LICENSE: WB-2348 | GST NUMBER: NOT SET | CONTACT: NOT SET | EMAIL: bose001@gmail.com

**Advanced Production Line**  
Auto-ID, Hierarchy & QR Generation

Select Product: Calpol (650 mg) - [Code: MED-83227588] | + Add New | Set MRP (Consumer Price): ₹ 100

**Packaging Configuration**

1. TOTAL CARTONS: 2 | CONTAINS: X | 2. BOXES PER CARTON: 4

3. STRIPS PER BOX: 10 | TOTAL PRODUCTION OUTPUT (Cartons x Boxes x Strips): 80 Strips

Manufacturing Date: 03/01/2025 | Expiry Date: 03/01/2026

**Confirm & Generate All QRs**

**Production Analytics**

Production Output (Last 7 Days): [Bar chart showing output for Fri]

THIS MONTH OUTPUT: 170 Strips

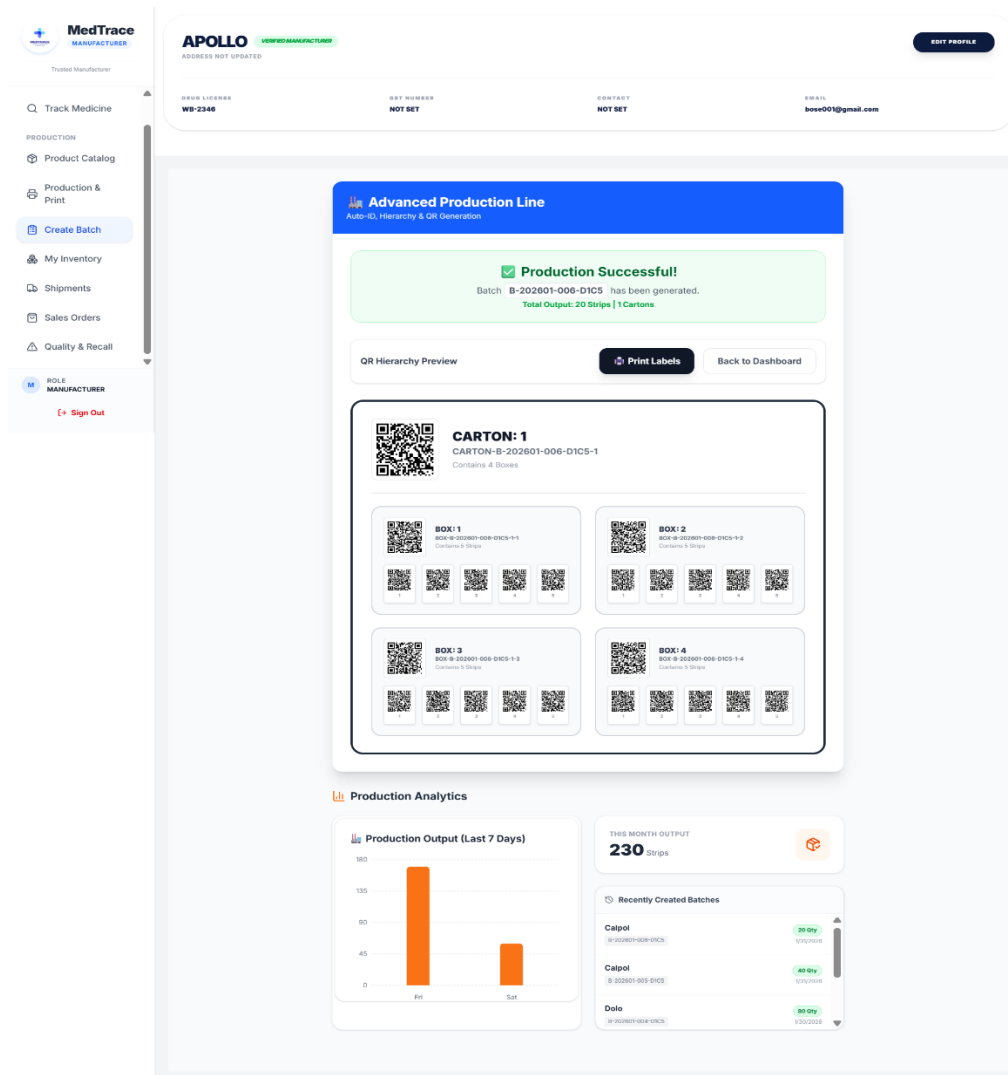
Recently Created Batches:

| Product         | QR Code           | Quantity | Date      |
|-----------------|-------------------|----------|-----------|
| Dolo            | 8-202401-004-01C3 | 80 Qty   | 1/30/2024 |
| Neomol Inj      | 8-202401-003-01C3 | 10 Qty   | 1/30/2024 |
| Diphenhydramine | 8-202401-002-01C3 | 40 Qty   | 1/30/2024 |

This page is used by the manufacturer to **create a new production batch and generate digital identities** for medicines. From this interface, the manufacturer selects the product and enters all required production details such as consumer price, packaging structure, manufacturing date, and expiry date.

The **packaging configuration** section allows the manufacturer to define the hierarchy of the batch, including the number of cartons, boxes per carton, and strips per box. Based on these inputs, the system automatically calculates the total production output and generates a unique digital identity for each unit.

After confirming the details, the manufacturer can click **“Confirm & Generate All QRs”** to automatically generate and assign secure QR codes for cartons, boxes, and individual strips. These codes are then printed and attached to the physical packages, enabling full traceability throughout the supply chain.



This page shows the **successful generation of a production batch and its QR code hierarchy** within the Digital Medicine Authentication and Traceability Framework. After the manufacturer confirms the batch details, the system automatically creates a unique batch ID and generates QR codes for each packaging level, including cartons, boxes, and individual strips.

The **QR hierarchy preview** visually displays the parent-child relationship between the carton, its boxes, and the strips contained inside each box. This ensures that all medicine units are digitally linked to their parent containers for efficient tracking.

From this screen, the manufacturer can **print labels** for all generated QR codes and attach them to the corresponding physical packages. These QR codes become the digital identities of the medicines and are used throughout the supply chain for scanning, tracking, and verification.

**MedTrace**  
MANUFACTURER  
Trusted Manufacturer

Track Medicine

PRODUCTION

- Product Catalog
- Production & Print
- Create Batch
- My Inventory**
- Shipments
- Sales Orders
- Quality & Recall

ROLE  
MANUFACTURER  
[Sign Out](#)

**APOLLO**  
VERIFIED MANUFACTURER  
ADDRESS NOT UPDATED

EDIT PROFILE

| DRUG LICENSE   | GST NUMBER     | CONTACT        | EMAIL             |
|----------------|----------------|----------------|-------------------|
| <b>WB-2346</b> | <b>NOT SET</b> | <b>NOT SET</b> | bose001@gmail.com |

### My Live Inventory

Real-time stock levels calculated from production and shipments.

LIVE UNITS  
**151**

CURRENT VALUE  
**₹15.4k**

| MEDICINE DETAILS                                              | BATCH INFO                                                 | PRODUCED          | LIVE STOCK                   | UNIT MRP | STATUS    |
|---------------------------------------------------------------|------------------------------------------------------------|-------------------|------------------------------|----------|-----------|
| <b>Calpol</b><br>Paracetamol<br><a href="#">TABLET</a>        | <b>B-202601-006-D1C5</b><br>Mfg: 2/1/2026<br>Exp: 2/1/2027 | 20<br>Total Batch | <b>20</b><br>Available Units | ₹100     | Low Stock |
| <b>Calpol</b><br>Paracetamol<br><a href="#">TABLET</a>        | <b>B-202601-005-D1C5</b><br>Mfg: 2/1/2026<br>Exp: 2/1/2027 | 40<br>Total Batch | <b>40</b><br>Available Units | ₹99.99   | Low Stock |
| <b>Dolo</b><br>Paracetamol<br><a href="#">TABLET</a>          | <b>B-202601-004-D1C5</b><br>Mfg: 2/2/2025<br>Exp: 2/2/2026 | 80<br>Total Batch | <b>69</b><br>Available Units | ₹99.98   | Expiring  |
| <b>Neomol Inj</b><br>Paracetamol<br><a href="#">INJECTION</a> | <b>B-202601-003-D1C5</b><br>Mfg: 2/1/2026<br>Exp: 2/1/2027 | 10<br>Total Batch | <b>2</b><br>Available Units  | ₹230     | Low Stock |
| <b>Calpol</b><br>Paracetamol<br><a href="#">TABLET</a>        | <b>B-202601-001-D1C5</b><br>Mfg: 3/2/2026<br>Exp: 3/2/2027 | 40<br>Total Batch | <b>20</b><br>Available Units | ₹99.96   | Low Stock |

This page represents the **Live Inventory Management screen** for the manufacturer. It displays real-time stock information that is automatically updated based on production and shipment records.

From this interface, the manufacturer can search and filter medicines or batches to quickly locate specific stock. Each entry shows essential details such as the medicine name, batch number, manufacturing date, expiry date, total produced units, and currently available stock.

The system also highlights critical conditions such as **low stock** and **expiring batches**, allowing the manufacturer to take timely action. This ensures that outdated or unsafe medicines can be identified early and removed from circulation if required.



**MedTrace**  
MANUFACTURER

Treated Manufacturer

Track Medicine

PRODUCTION

Product Catalog

Production & Print

Create Batch

My Inventory

Shipments

Sales Orders

Quality & Recall

M

ROLE  
MANUFACTURER

Sign Out

**APOLLO**
VERIFIED MANUFACTURER

ADDRESS NOT UPDATED

EDIT PROFILE

DRUG LICENSE  
**WB-2346**

GST NUMBER  
**NOT SET**

CONTACT  
**NOT SET**

EMAIL  
**bose001@gmail.com**

**Dispatch & Logistics**  
Create shipments, generate invoices, and track logistics.

1. SELECT DISTRIBUTOR

saptaparni  
LIC: MH-20-12345

shreshtha  
LIC: 10/GIA/2013

Rohit  
LIC: MH- 23570

Roy  
LIC: DL-9089

Saha  
LIC: GJ-8647

INVOICESUMMARY

Cart is empty

PREVIEW INVOICE

CONFIRM & DISPATCH

\*Clicking confirm will update stock in database.

2. ADD ITEMS

| BATCH ID    | PRODUCT    | STOCK    | ACTION |
|-------------|------------|----------|--------|
| 245e41ef... | Calpol     | 40 Units | + ADD  |
| 614d6150... | Calpol     | 20 Units | + ADD  |
| 152f725a... | Dolo       | 70 Units | + ADD  |
| 9579bd54... | Calpol     | 20 Units | + ADD  |
| 86e237cc... | Neomal Ini | 5 Units  | + ADD  |

Logistics Overview

Delivery Status

IN TRANSIT: 1   DELIVERED: 4

DISPATCHED VALUE (THIS MONTH)

₹8,957

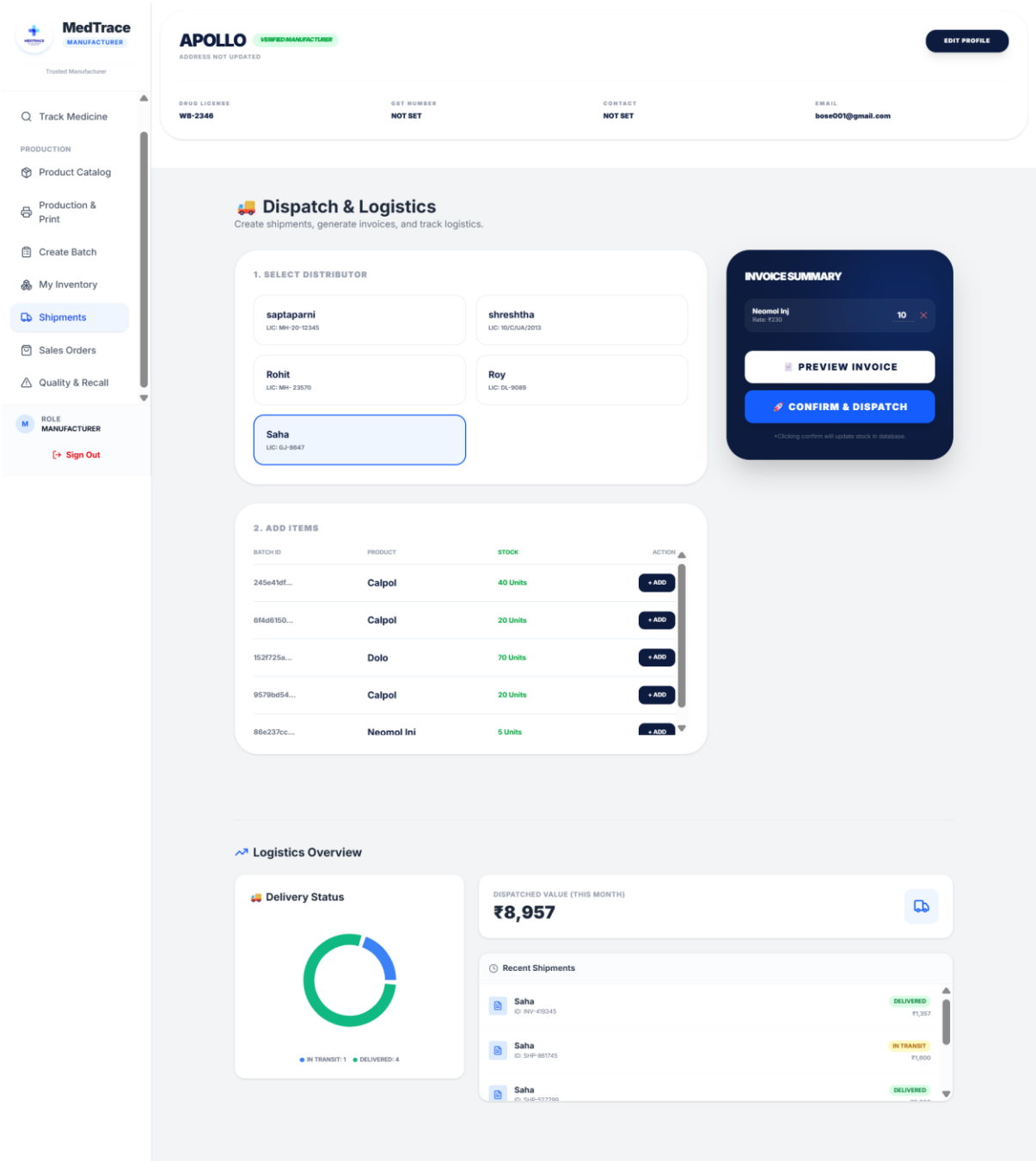
Recent Shipments

|      |                |            |        |
|------|----------------|------------|--------|
| Saha | ID: INV-498345 | DELIVERED  | ₹1,357 |
| Saha | ID: SHP-881743 | IN TRANSIT | ₹1,600 |
| Saha | ID: SHP-537796 | DELIVERED  | ₹1,600 |

This page represents the **Dispatch and Logistics module** of the Digital Medicine Authentication and Traceability Framework. From this interface, the manufacturer can select a distributor, choose medicine batches from available stock, and add them to a shipment. The system allows the user to generate and preview the invoice before confirming the dispatch. Once the shipment is confirmed, the stock database is automatically updated and the ownership of the selected medicine units is transferred to the distributor.

The lower section of the page provides a **logistics overview**. The delivery status chart visually shows the number of shipments that are currently in

transit and those that have already been delivered. This helps the manufacturer quickly understand the progress of ongoing shipments. The **recent shipments** panel displays a list of the latest dispatch records, along with their delivery status such as “Delivered” or “In Transit.” This section allows the manufacturer to track shipment history and verify that medicines are moving through the supply chain as expected.



**This figure shows** the Dispatch & Logistics page of the MedTrace Manufacturer system, where a shipment of injection medicine is being created for the distributor **Saha**, including product selection, invoice generation, and dispatch confirmation.

# Apollo

Industrial Estate, India  
GSTIN: GST-PENDING | DL: WB-2346  
Email: support@medtrace.com

## INVOICE

NO: INV-306718  
Date: 31/01/2026  
Status: Unpaid

### BILLED TO:

Saha  
South Kolkata, Jadavpur  
GSTIN: N/A

| # | Medicine                        | Batch  | Qty | Price         | Tax % | Tax Amt       | Total          |
|---|---------------------------------|--------|-----|---------------|-------|---------------|----------------|
| 1 | Neomol Inj<br>(Batch: 86e237cc) | 86E237 | 10  | INR<br>230.00 | 18%   | INR<br>414.00 | INR<br>2714.00 |

### Bank Details:

Bank: HDFC Bank  
A/C No: XXXXXXXXXX5678  
IFSC: HDFC0001234

Sub Total: INR 2300.00  
Tax (18%): INR 414.00

**Grand Total: INR 2714.00**

### Terms & Conditions:

- Goods once sold will not be taken back.
- Interest @18% p.a. will be charged if not paid within due date.

For Apollo

Authorized Signatory

**This figure shows** the generated invoice for the shipment of injection medicine from Apollo to the distributor **Saha**. It includes the medicine name (Neomol Inj), batch number, quantity, price, tax details, and the grand total amount, along with billing and bank information.

**MedTrace**  
MANUFACTURER  
Trusted Manufacturer

Dashboard

Track Medicine

PRODUCTION

Product Catalog

Production & Print

Create Batch

My Inventory

Shipments

Sales Orders

ROLE MANUFACTURER

Sign Out

**APOLLO**  
ADDRESS NOT UPDATED  
VERIFIED MANUFACTURER

DRUG LICENSE  
**WB-2346**

GST NUMBER  
**NOT SET**

CONTACT  
**NOT SET**

EMAIL  
**bose001@gmail.com**

EDIT PROFILE

Incoming Orders

Manage B2B orders received from distributors.

5 Orders

#ORD-661891

PENDING

TOTAL VALUE ₹800

Ordered By: Roy

1/31/2026

| Product Name | Quantity | Price/Unit | Total |
|--------------|----------|------------|-------|
| Calpol       | 10       | ₹80        | ₹800  |

Reject

Approve Order

#ORD-273915

SHIPPED

TOTAL VALUE ₹2,600

Ordered By: Roy

1/30/2026

| Product Name    | Quantity | Price/Unit | Total |
|-----------------|----------|------------|-------|
| Diphenhydramine | 20       | ₹130       | ₹2600 |

#ORD-035274

SHIPPED

TOTAL VALUE ₹2,600

Ordered By: Saha

1/30/2026

| Product Name    | Quantity | Price/Unit | Total |
|-----------------|----------|------------|-------|
| Diphenhydramine | 20       | ₹130       | ₹2600 |

#ORD-989050

SHIPPED

TOTAL VALUE ₹800

Ordered By: Saha

1/30/2026

| Product Name | Quantity | Price/Unit | Total |
|--------------|----------|------------|-------|
| Dolo         | 10       | ₹80        | ₹800  |

#ORD-821123

SHIPPED

TOTAL VALUE ₹1,600

Ordered By: Saha

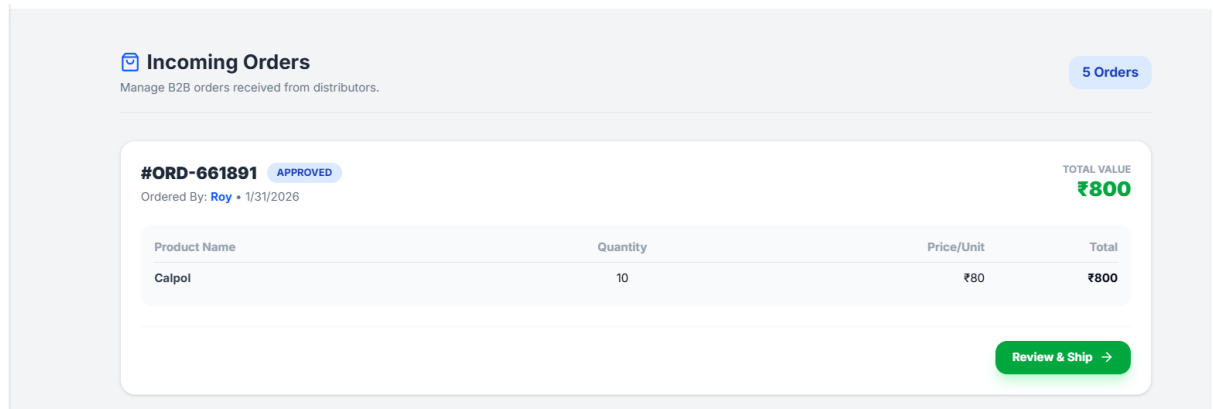
1/30/2026

| Product Name | Quantity | Price/Unit | Total |
|--------------|----------|------------|-------|
| Calpol       | 20       | ₹80        | ₹1600 |

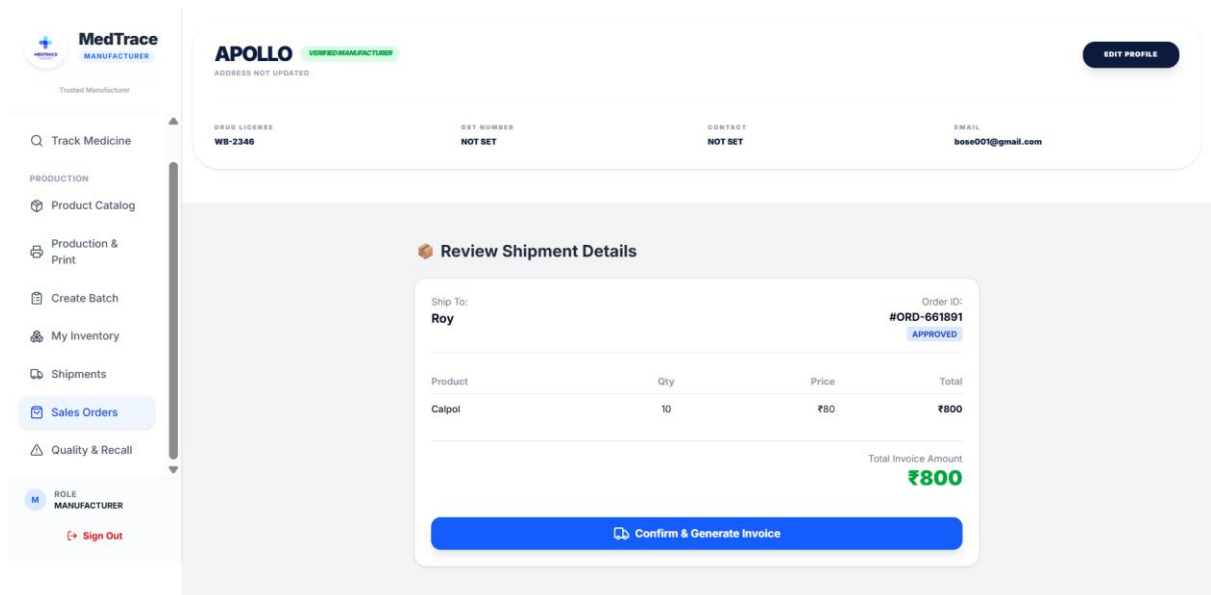
This page represents the **Incoming Sales Orders module** of the Digital Medicine Authentication and Traceability Framework. It displays all purchase orders placed by distributors to the manufacturer.

Each order shows the order ID, distributor name, order date, product details, quantity, and current status such as *Pending* or *Shipped*. For pending requests, the manufacturer can **approve or reject** the order directly from this page.

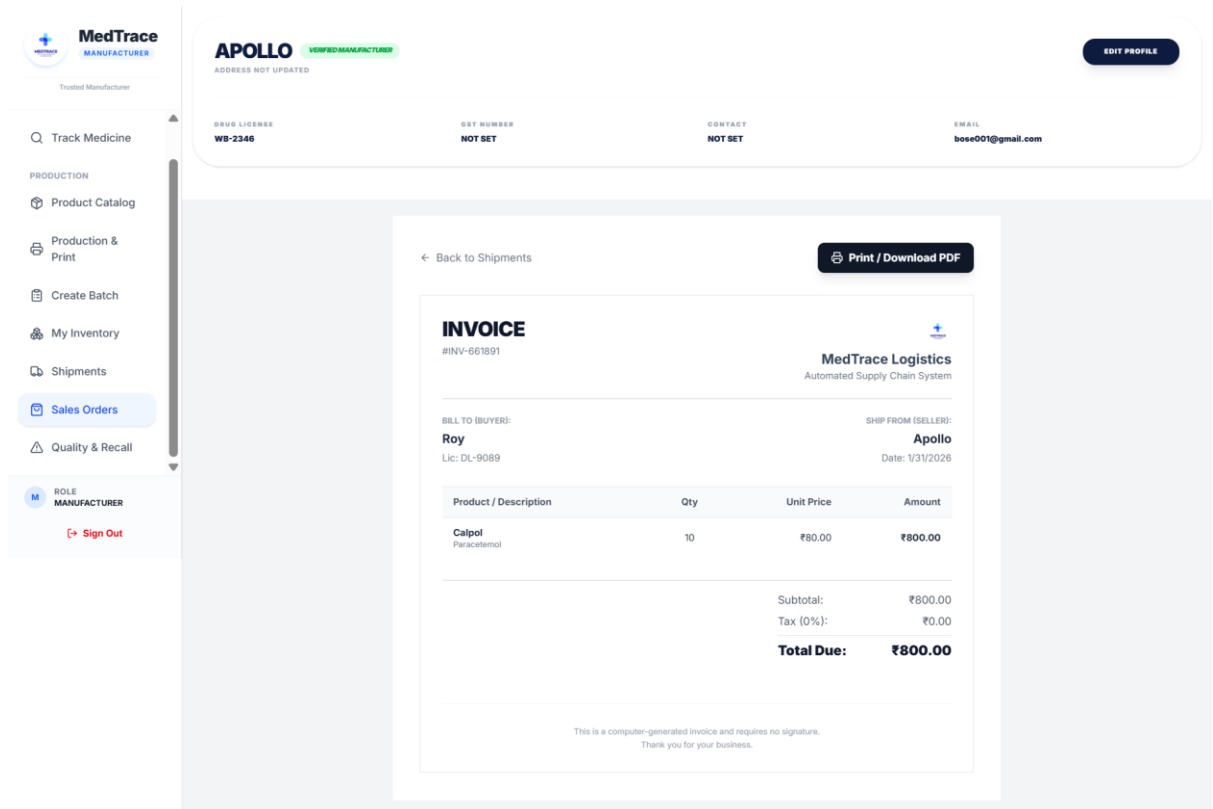
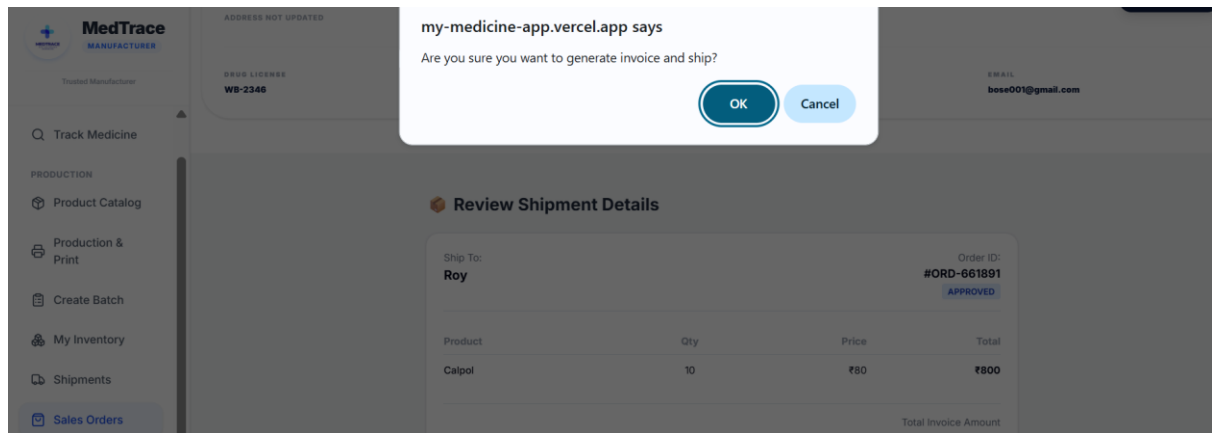
When an order is approved, the system automatically **generates the invoice**, updates inventory, and initiates the shipment process to the distributor. If an order is rejected, the distributor is notified, and the request is closed.



After approving the order, the “Review & Ship” option appears, allowing the manufacturer to verify the order details and proceed with dispatch to the distributor.



**This figure shows** the *Review Shipment Details* page, where the manufacturer verifies the approved order from **Roy**, checks the product, quantity, price, and total amount, and then clicks **Confirm & Generate Invoice** to proceed with the shipment.



**This figure shows** the final generated invoice in the MedTrace system for the approved order from distributor **Roy**, including product details, quantity, price, subtotal, and total due, with options to print or download the invoice as a PDF.



MedTrace  
MANUFACTURER

Treated Manufacturer

**APOLLO** VERIFIED MANUFACTURER

ADDRESS NOT UPDATED

EDIT PROFILE

DRUG LICENSE: WB-2346

SET NUMBER: NOT SET

CONTACT: NOT SET

EMAIL: bese001@gmail.com

Track Medicine

PRODUCTION

Product Catalog

Production & Print

Create Batch

My Inventory

Shipments

Sales Orders

**Quality & Recall**

ROLE: MANUFACTURER

[Sign Out](#)

### Quality Control

Monitor batch quality and issue emergency recalls.

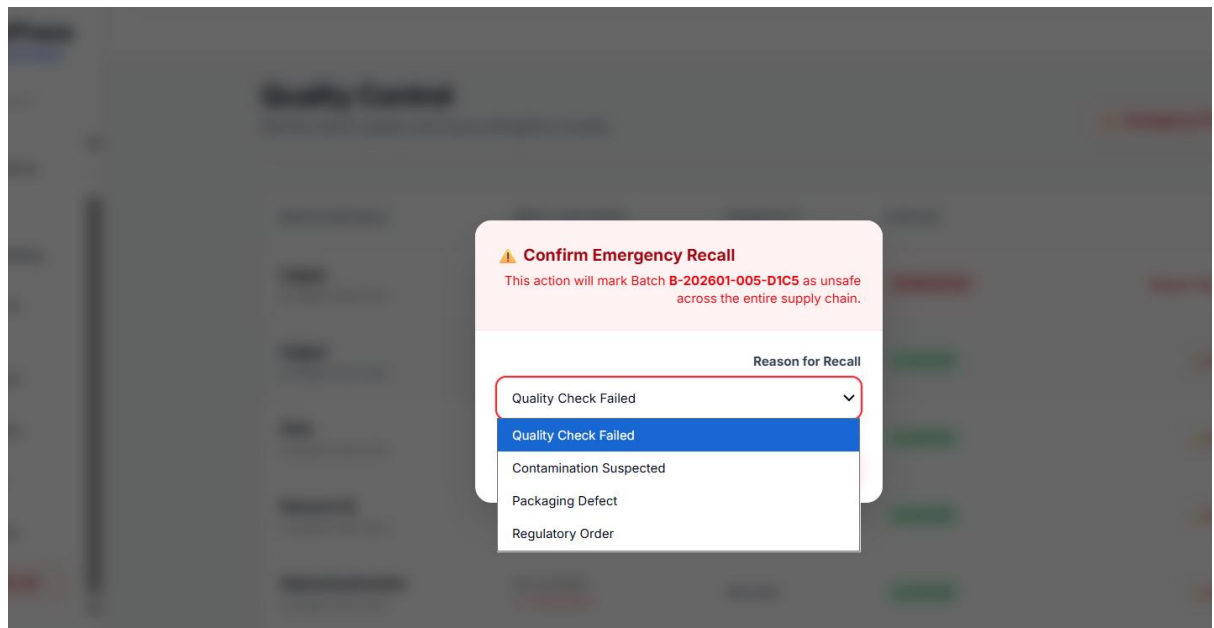
**Emergency Protocol Active**

| BATCH DETAILS                               | MFG / EXP DATE              | QUANTITY | STATUS          | ACTION                       |
|---------------------------------------------|-----------------------------|----------|-----------------|------------------------------|
| <b>Calpol</b><br>B-202601-006-DIC5          | M: 2/1/2026<br>E: 2/1/2027  | 20 Units | <b>RECALLED</b> | Reason: Packaging Defect     |
| <b>Calpol</b><br>B-202601-005-DIC5          | M: 2/1/2026<br>E: 2/1/2027  | 40 Units | <b>ACTIVE</b>   | <a href="#">Recall Batch</a> |
| <b>Dolo</b><br>B-202601-004-DIC5            | M: 2/2/2025<br>E: 2/2/2026  | 80 Units | <b>ACTIVE</b>   | <a href="#">Recall Batch</a> |
| <b>Neomol Inj</b><br>B-202601-003-DIC5      | M: 2/1/2026<br>E: 2/1/2027  | 10 Units | <b>ACTIVE</b>   | <a href="#">Recall Batch</a> |
| <b>Diphenhydramine</b><br>B-202601-002-DIC5 | M: 2/1/2025<br>E: 1/20/2026 | 40 Units | <b>ACTIVE</b>   | <a href="#">Recall Batch</a> |
| <b>Calpol</b><br>B-202601-001-DIC5          | M: 3/2/2026<br>E: 3/2/2027  | 40 Units | <b>ACTIVE</b>   | <a href="#">Recall Batch</a> |

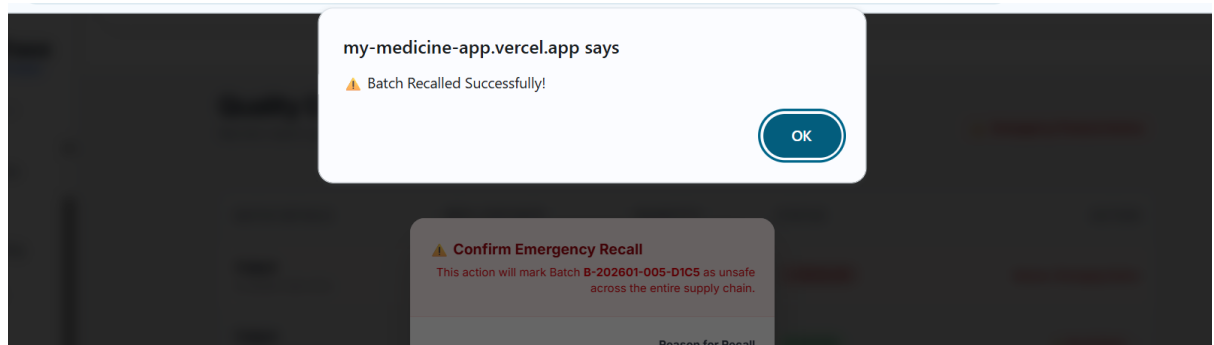
This page represents the **Quality Control and Recall Management module**, which is one of the most unique and critical features of the Digital Medicine Authentication and Traceability Framework. From this interface, the manufacturer can monitor the quality status of every batch and **initiate an emergency recall** if a defect, contamination, or safety issue is detected.

When a batch is marked as **recalled**, the system instantly updates its status in the central database. Because every unit is digitally linked to its movement history, **all distributors and retailers who currently hold the affected medicines automatically receive a notification**. The recalled status is also shown immediately on the verification interface whenever the medicine is scanned.

This real-time recall mechanism enables the **immediate stopping of medicine distribution and sale**, preventing unsafe products from reaching consumers. Unlike traditional recall systems that rely on slow manual communication, this automated digital recall process ensures faster response, improved accountability, and enhanced patient safety.



**This figure shows** the emergency recall confirmation window, where the manufacturer selects a specific reason for recalling a medicine batch—such as *Quality Check Failed*, *Contamination Suspected*, *Packaging Defect*, or *Regulatory Order*—before marking the batch as unsafe across the entire supply chain.



This figure shows that the batch has been successfully recalled.

- **Consumer/User interface**

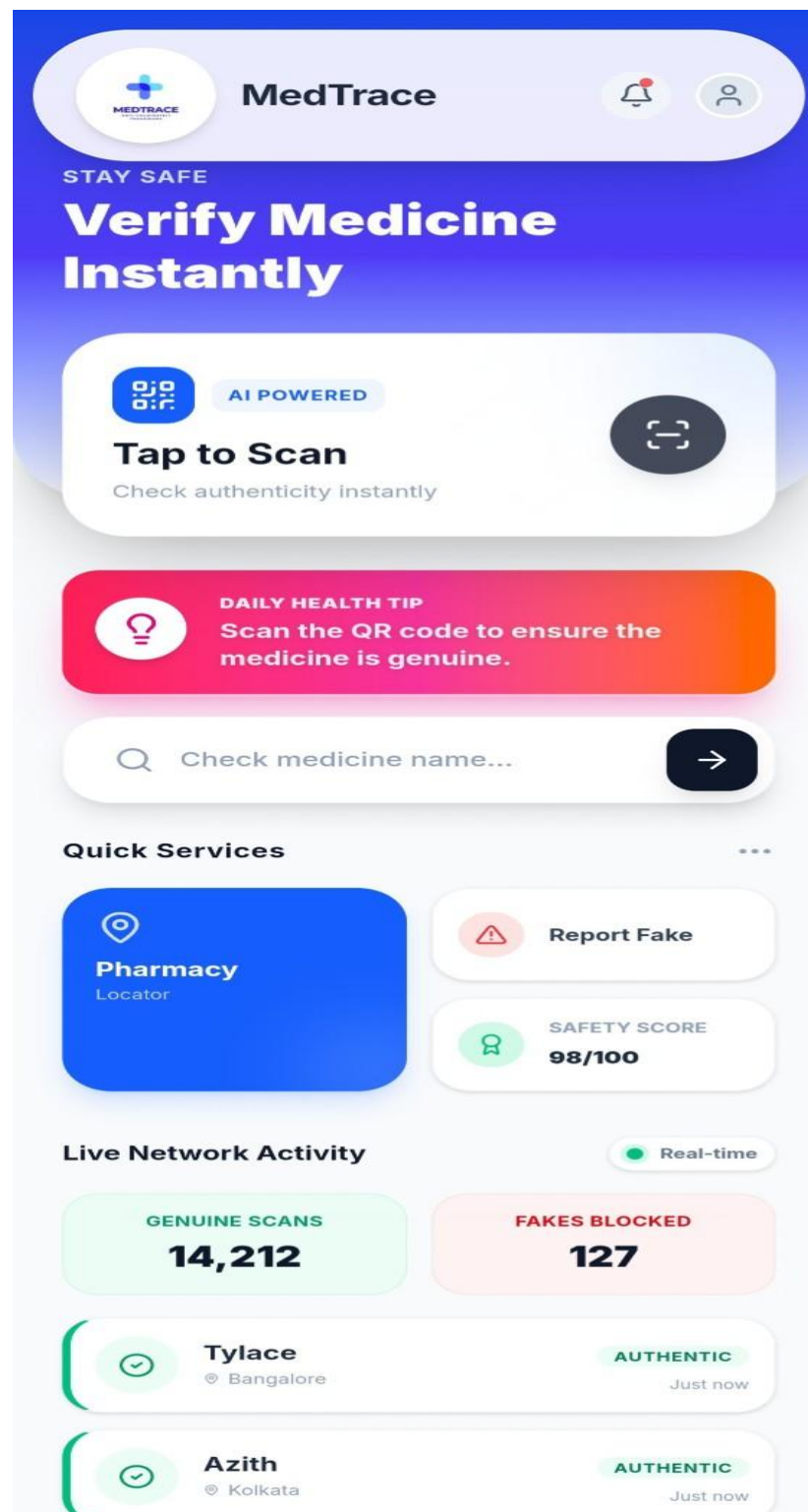
This figure shows the mobile landing page of the Digital Medicine Authentication and Traceability Framework. It is the user-facing interface that allows consumers, pharmacists, and field inspectors to verify medicines instantly.

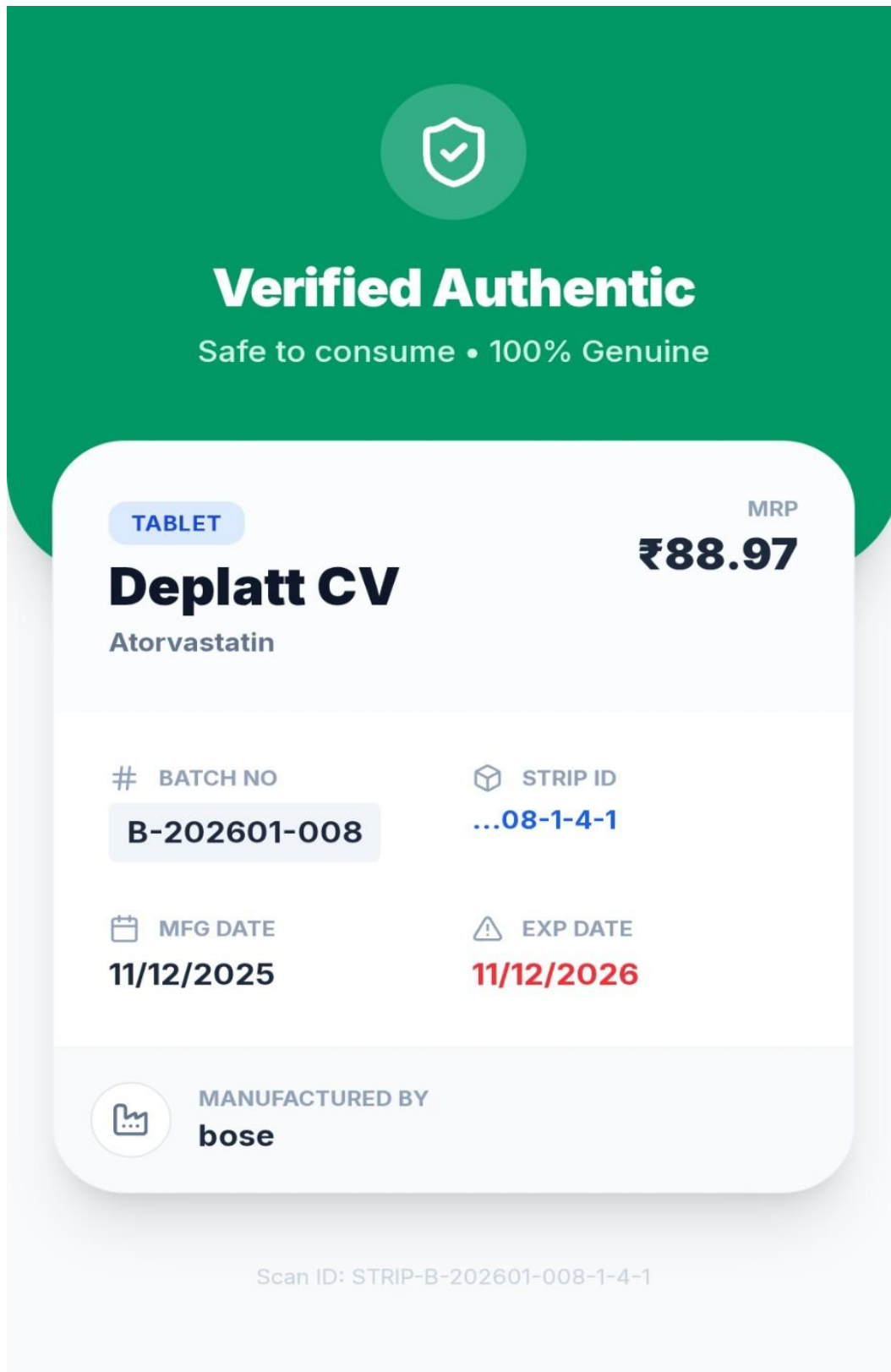
From this screen, users can tap the “Scan” option to scan the QR code printed on the medicine package. The scanned code is sent to the central verification server, which checks the medicine’s authenticity, status, and



safety information in real time. The result is then displayed on the same screen.

The page also provides quick services such as pharmacy locator, fake medicine reporting, and a visible safety score. Live network activity shows the number of genuine scans and blocked fake attempts, helping build user trust.





This figure shows the **successful verification result screen** displayed after scanning a valid medicine using the platform verification interface. The green background and shield icon clearly indicate that the medicine is **verified, authentic, and safe to consume**.

The screen presents complete and transparent information about the medicine, including the medicine name, generic name, dosage form, batch number, and unique strip identifier. Manufacturing and expiry dates are prominently displayed so that users can easily check product validity. The manufacturer name and MRP are also shown to ensure pricing transparency.



This figure represents the **recall alert screen** that is displayed when a user scans a medicine that has been officially recalled by the manufacturer. The warning interface uses a red color theme and alert icon to immediately draw attention and indicate danger. It clearly states that the product batch has been recalled due to safety concerns and instructs the user **not to consume the medicine**.

The screen also displays the **medicine name and batch number**, allowing the user to quickly identify the affected product. This ensures

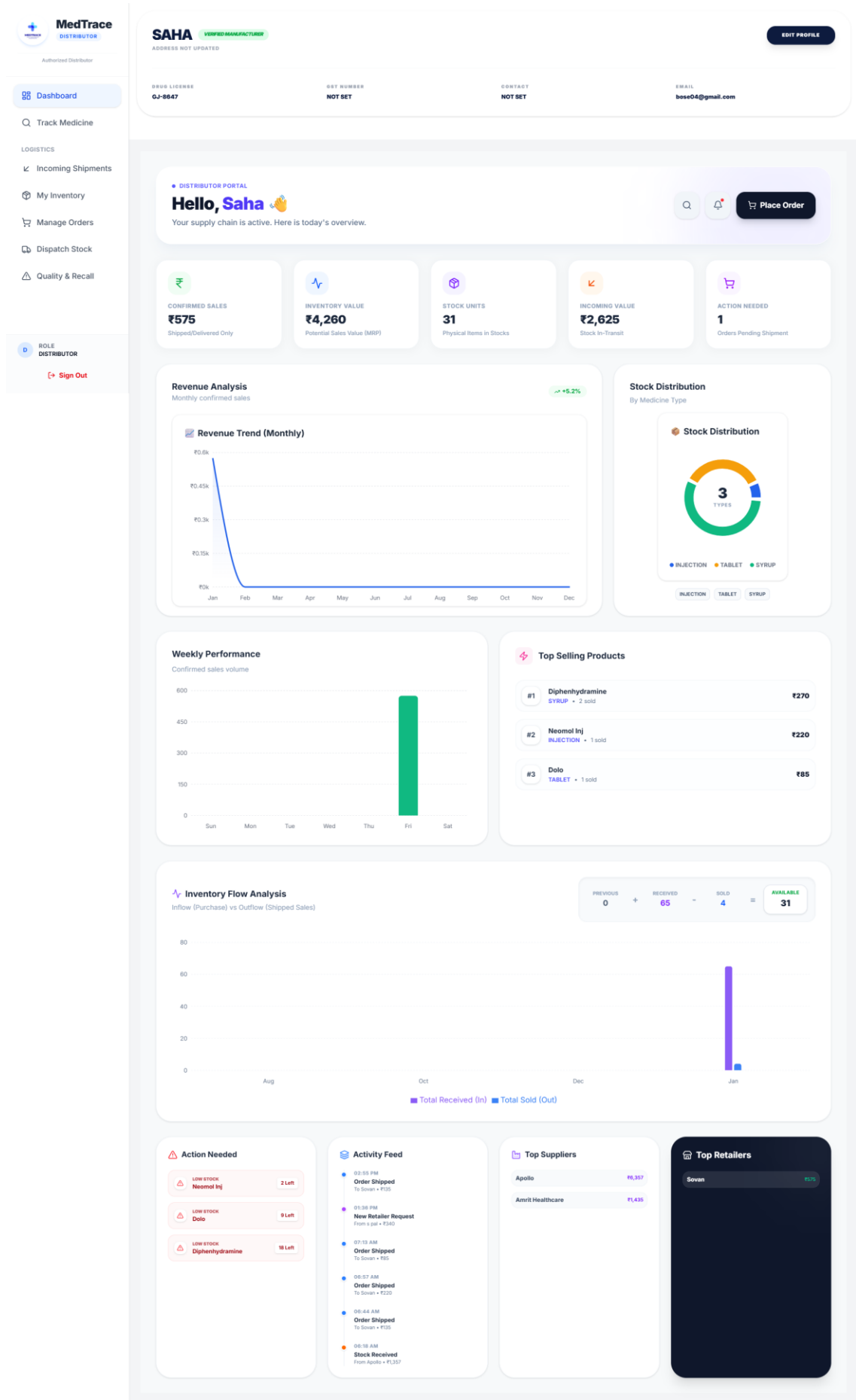
transparency and prevents confusion in case of large-scale recalls involving multiple medicines.

This feature is one of the most critical and unique aspects of the system. Once a manufacturer marks a batch as recalled in the Quality Control module, the recall status is instantly updated across the entire network. All distributors, retailers, and consumers who scan the affected medicine receive this warning in real time.

As a result, the system enables **immediate supply chain shutdown for defective batches**, helping to prevent health risks, reduce harm, and ensure patient safety.

## **2. Distributor Login Page**

The figure shows the **Distributor Dashboard** of the MedTrace system. After login, the distributor can view real-time information about sales, stock, and incoming medicines. The dashboard displays total inventory, stock units, confirmed sales, and pending actions in one place.



It also shows graphical reports such as monthly sales trends, weekly performance, and stock distribution by medicine type. The distributor can easily see top-selling products, low-stock alerts, recent activities, and details of suppliers and retailers.

Overall, this page helps the distributor track inventory, manage orders, and monitor medicine movement efficiently within the supply chain.

**MedTrace**  
DISTRIBUTOR

Authorized Distributor

Dashboard

Track Medicine

LOGISTICS

Incoming Shipments

My Inventory

Manage Orders

Dispatch Stock

Quality & Recall

ROLE DISTRIBUTOR

Sign Out

**SAHA**  
VERIFIED MANUFACTURER  
ADDRESS NOT UPDATED

EDIT PROFILE

DRUG LICENSE

GST NUMBER

CONTACT

EMAIL

GJ-8647

NOT SET

NOT SET

bose04@gmail.com

**B2B Marketplace**  
Browse catalogs from manufacturers and restock your inventory.

**Calpol**  
Paracetamol  
TABLET

Amrit Healthcare

Price: ₹82 / unit

Strength: 650

100

Order

**Calpol**  
Paracetamol  
TABLET

Apollo

Price: ₹80 / unit

Strength: 650 mg

100

Order

**Cipolex**  
Eye drops  
DROPS

Apollo

Price: ₹15 / unit

Strength: 0.3%

100

Order

**DOLO**  
Paracetamol  
TABLET

bose

Price: ₹80 / unit

Strength: 650mg

100

Order

**Diphenhydramine**  
Benadryl  
SYRUP

Apollo

Price: ₹130 / unit

Strength: 150 mg

100

Order

**Diphenhydramine**  
Benadryl  
SYRUP

Amrit Healthcare

Price: ₹135 / unit

Strength: 200 mg

100

Order

**Dolo**  
Paracetamol  
TABLET

Amrit Healthcare

Price: ₹85 / unit

Strength: 650

100

Order

**Dolo**  
Paracetamol  
TABLET

Apollo

Price: ₹80 / unit

Strength: 650 mg

100

Order

**Gaspaz**  
Domperidone and Omeprazole  
CAPSULE

bose

Price: ₹180 / unit

Strength: 500 mg

100

Order

**Neomol Inj**  
Paracetamol  
INJECTION

Amrit Healthcare

Price: ₹205 / unit

Strength: 200 mg

100

Order

**Neomol Inj**  
Paracetamol  
INJECTION

Apollo

Price: ₹210 / unit

Strength: 1 g/100 ml

100

Order

**Volini**  
Diclofenac Sodium  
SPRAY

Apollo

Price: ₹60 / unit

Strength: 75 mg

100

Order

**Zelocip CF Cream**  
Flucinolone  
CREAM

Apollo

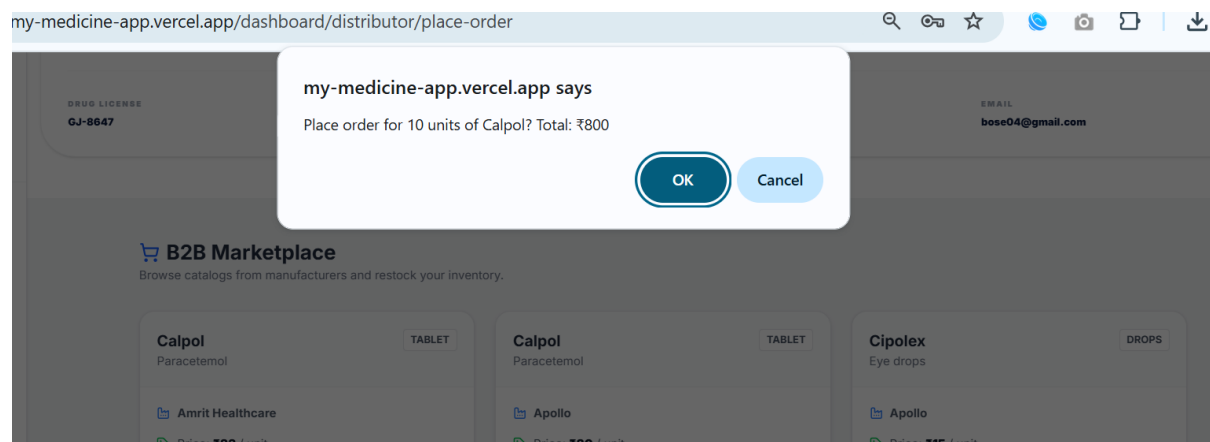
Price: ₹35 / unit

Strength: 25 g

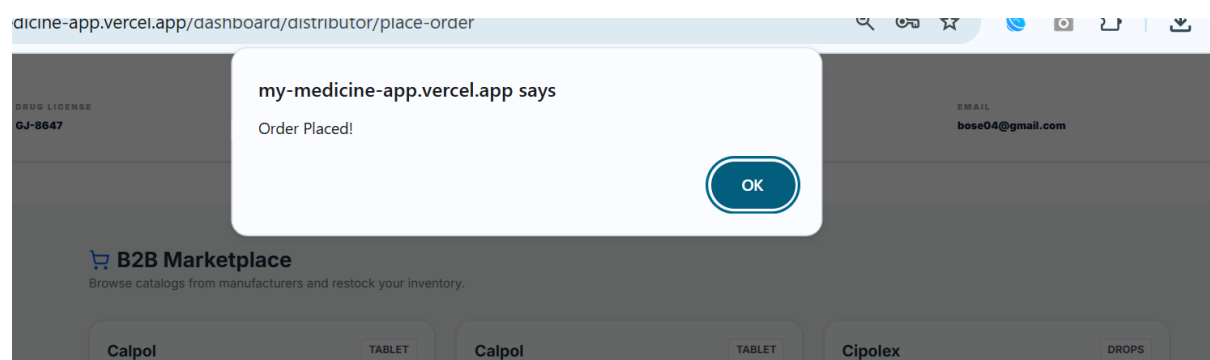
100

Order

**This figure shows** the *B2B Marketplace* page in the MedTrace distributor panel, where the distributor can browse medicines from different manufacturers, view price, strength, and type, enter required quantity, and place orders directly to restock their inventory.



**This figure shows** the order confirmation popup in the MedTrace B2B Marketplace, where the distributor is asked to confirm placing an order for **10 units of Calpol** with a total amount of **₹800** before proceeding.



**This figure shows** a confirmation message indicating that the distributor's order has been **successfully placed** in the MedTrace B2B Marketplace system.

**MedTrace**  
DISTRIBUTOR

Authorized Distributor

Dashboard

Track Medicine

LOGISTICS

Incoming Shipments

My Inventory

Manage Orders

Dispatch Stock

Quality & Recall

ROLE DISTRIBUTOR

Sign Out

SAHA

VERIFIED MANUFACTURER

ADDRESS NOT UPDATED

EDIT PROFILE

DRUG LICENSE

6J-8647

SET NUMBER

NOT SET

CONTACT

NOT SET

EMAIL

bose04@gmail.com

Incoming Shipments

Total Pending: 2

Review and accept stock from manufacturers.

SHIPMENT ID

SHP-861745

07375839-4940-4309-9788-3d4d42f896d8

SENDER (MANUFACTURER)

Apollo

TOTAL VALUE

₹1,600

DATE

1/30/2026

SHIPMENT CONTENTS:

| MEDICINE NAME                                   | BATCH DETAILS (NO & UUID)                                                      | QTY       | PRICE |
|-------------------------------------------------|--------------------------------------------------------------------------------|-----------|-------|
| <div>Calpol</div> <div>Code: MED-83227588</div> | <div>B-202601-001-D1C5</div> <div>3f707b03d4-8ac0-4805-a236-4c7a5c757788</div> | 20 strips | ₹80   |

Receive & Add to Stock

SHIPMENT ID

SHP-659438

64375776-7876-4a4a-4650-9d48267048d8

SENDER (MANUFACTURER)

Amrit Healthcare

TOTAL VALUE

₹1,025

DATE

1/30/2026

SHIPMENT CONTENTS:

| MEDICINE NAME                                       | BATCH DETAILS (NO & UUID)                                                    | QTY      | PRICE |
|-----------------------------------------------------|------------------------------------------------------------------------------|----------|-------|
| <div>Neomol Inj</div> <div>Code: MED-35640222</div> | <div>B-202601-001-6AE2</div> <div>37868058-8c49-484b-8967-4c7a5c757788</div> | 5 strips | ₹205  |

Receive & Add to Stock

This page shows the **Incoming Shipments** section of the Distributor portal. From here, the distributor can review all medicines sent by manufacturers, along with shipment ID, sender name, batch details, quantity, price, and date.

Each shipment has a **“Receive & Add to Stock”** button. When the distributor clicks it, the system confirms the delivery and automatically updates the distributor’s inventory. This ensures that every medicine unit is digitally tracked as soon as it enters the distributor’s stock.

MedTrace

DISTRIBUTOR

Authorized Distributor

Dashboard

Track Medicine

my-medicine-app.vercel.app says

Are you sure you received this shipment physically?

OK

Cancel

Incom

Review and

SHIPMENT ID

INV-52

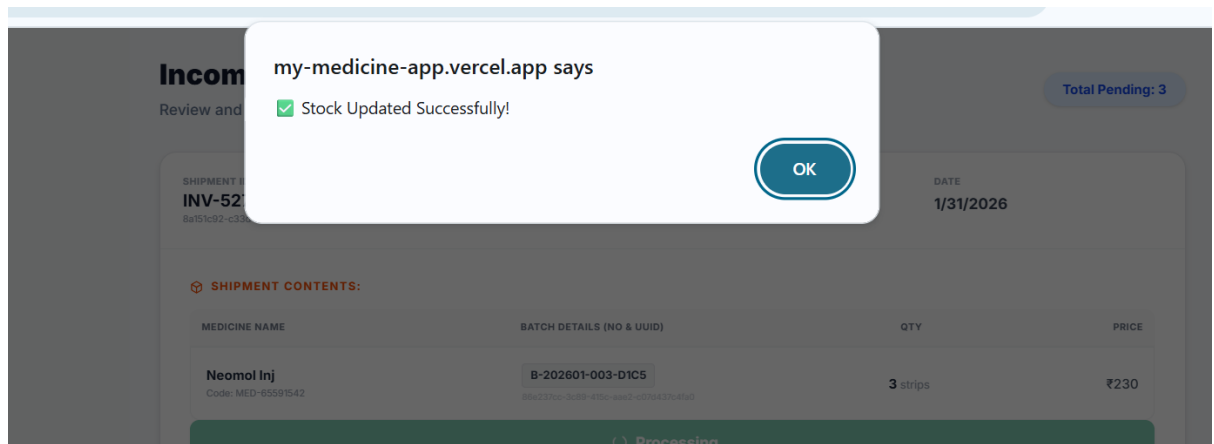
8a751c92-c3a5-4309-9788-3d4d42f896d8

DATE

1/31/2026

SHIPMENT CONTENTS:





This figure shows that the stock successfully updated to the database.

**MedTrace**  
DISTRIBUTOR  
Authorized Distributor

**SAHA** VERIFIED MANUFACTURER  
ADDRESS NOT UPDATED

EDIT PROFILE

DRUG LICENSE: GJ-9647 | GST NUMBER: NOT SET | CONTACT: NOT SET | EMAIL: boso04@gmail.com

**My Inventory**  
Manage stock and set selling prices for retailers.

TOTAL INVENTORY VALUE: ₹4.3k (Real-time Valuation)

STOCK COMPOSITION: Donut chart showing TABLET, SYRUP, and INJECTION.

EXPIRY RISK ANALYSIS: Bar chart showing Safe (0-6m), Warning (3-6m), and Critical (<3m) segments.

4 ACTIVE BATCHES | 4 LOW STOCK ALERTS

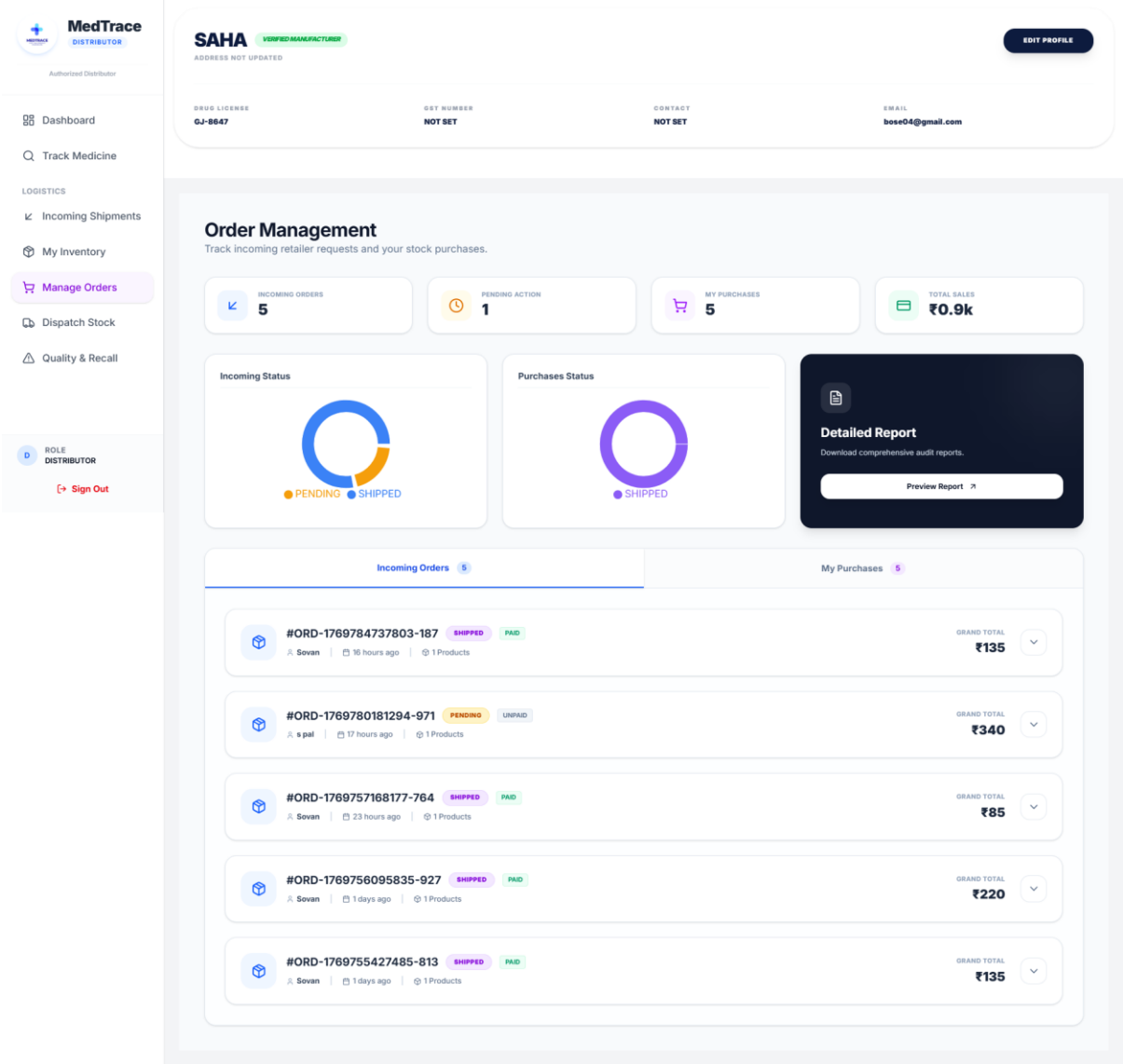
Search medicine or batch... Filter

| MEDICINE NAME                                 | BATCH DETAILS                              | STOCK LEVEL | BUYING PRICE (M) | MRP    | YOUR SELLING PRICE |
|-----------------------------------------------|--------------------------------------------|-------------|------------------|--------|--------------------|
| <b>Calpol</b><br>Paracetamol<br>TABLET        | B-202601-002-6AE2<br>Mfr: Anrit Healthcare | 2 Units     | ₹82              | ₹100   | ₹ 90               |
| <b>Diphenhydramine</b><br>Benadryl<br>SYRUP   | B-202601-002-D1C5<br>Mfr: Apollo           | 18 Units    | ₹130             | ₹150   | ₹ 135              |
| <b>Dolo</b><br>Paracetamol<br>TABLET          | B-202601-004-D1C5<br>Mfr: Apollo           | 9 Units     | ₹80              | ₹99.98 | ₹ 85               |
| <b>Neomol Inj</b><br>Paracetamol<br>INJECTION | B-202601-003-D1C5<br>Mfr: Apollo           | 2 Units     | ₹210             | ₹230   | ₹ 220              |

ROLE: DISTRIBUTOR | Sign Out

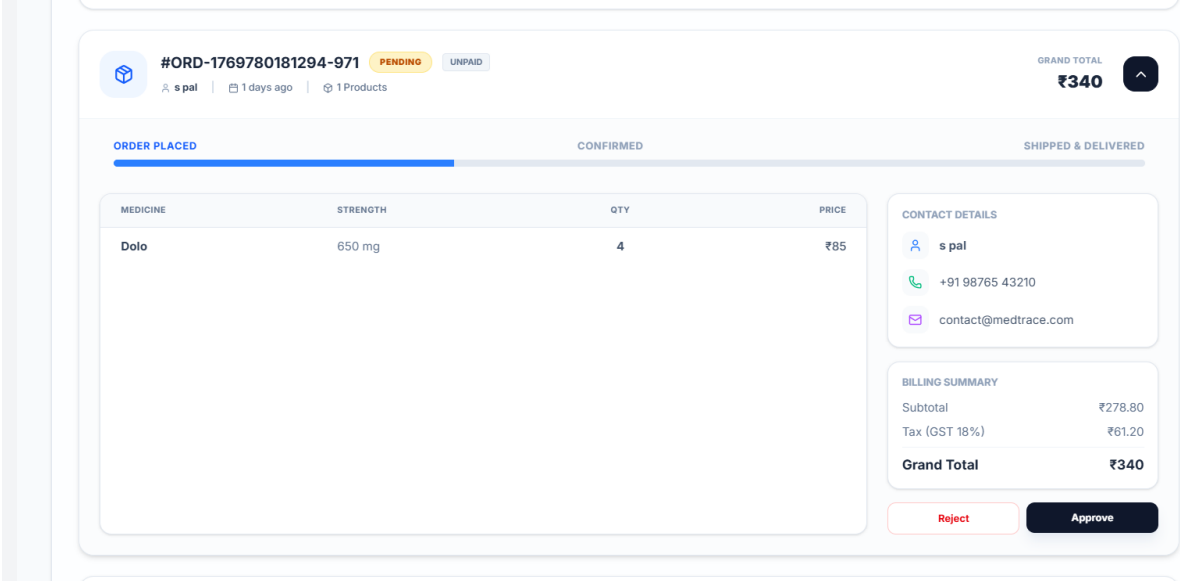
This page represents the **My Inventory** section of the Distributor portal. It allows the distributor to view and manage all medicines currently in stock in real time. The dashboard shows total inventory value, number of

active batches, low stock alerts, stock composition by medicine type, and expiry risk analysis based on remaining shelf life. Below, a detailed table lists each medicine with its batch number, stock level, buying price, MRP, and the distributor's own selling price. From this screen, the distributor can quickly identify low-stock or near-expiry items and adjust prices for retailers, ensuring efficient stock management and uninterrupted supply.

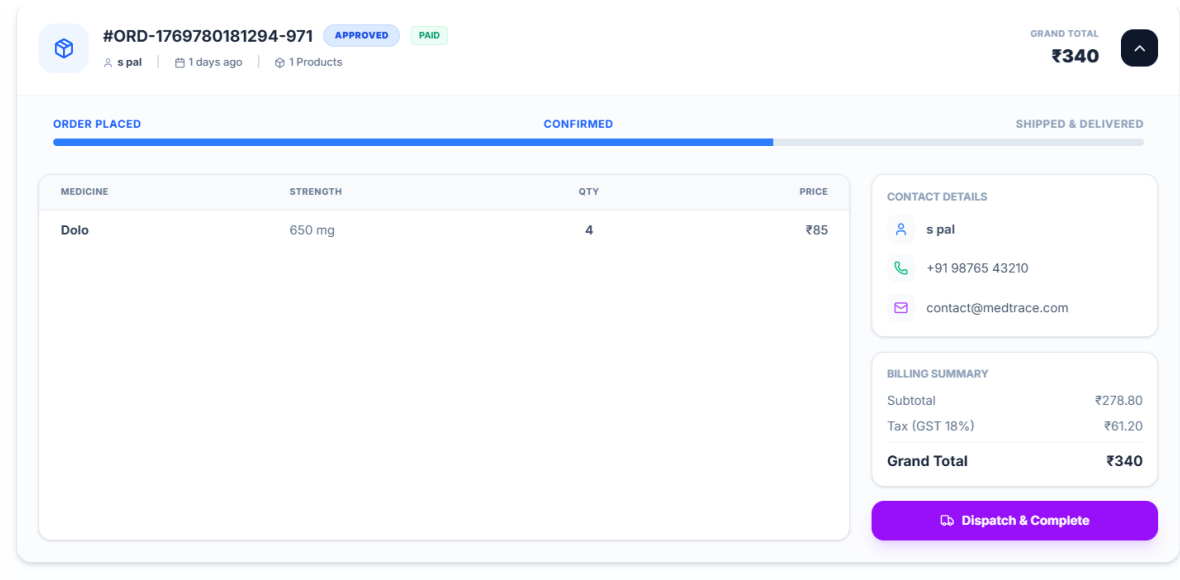


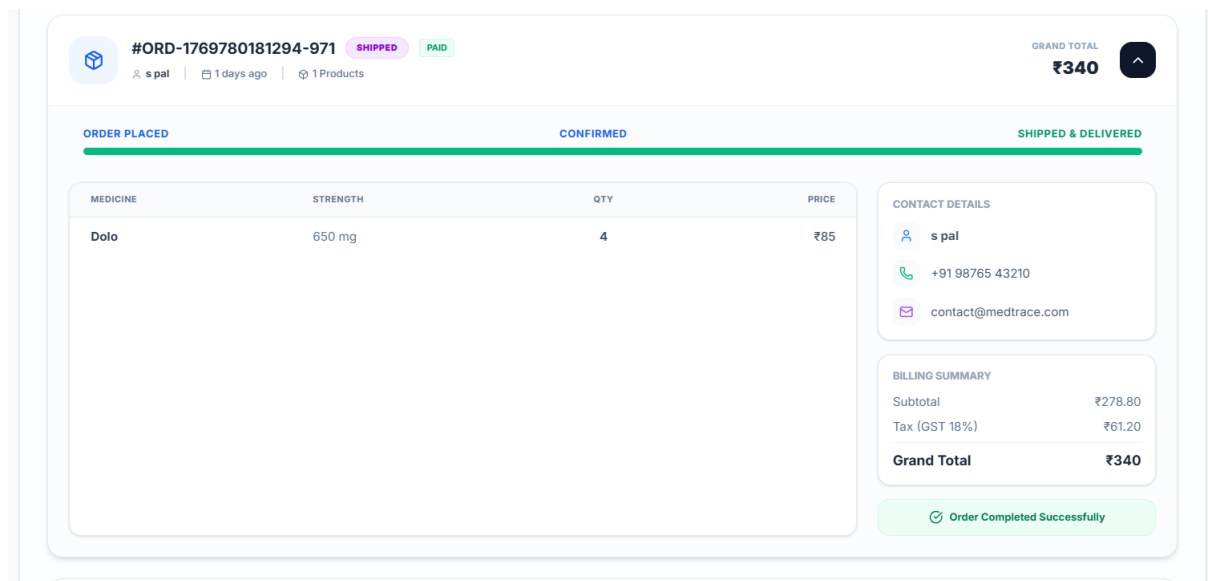
This page is the **Order Management** section of the Distributor portal. It allows the distributor to track all incoming retailer orders as well as their own purchase records in one place. The top summary shows the total number of incoming orders, pending actions, completed purchases, and overall sales value. The circular charts give a quick visual view of order status such as **pending** and **shipped**, while the order list below displays each order with its ID, customer name, time, payment status, shipping status, and total

amount. From this interface, the distributor can monitor order flow, take action on pending requests, and maintain smooth coordination with retailers and manufacturers.

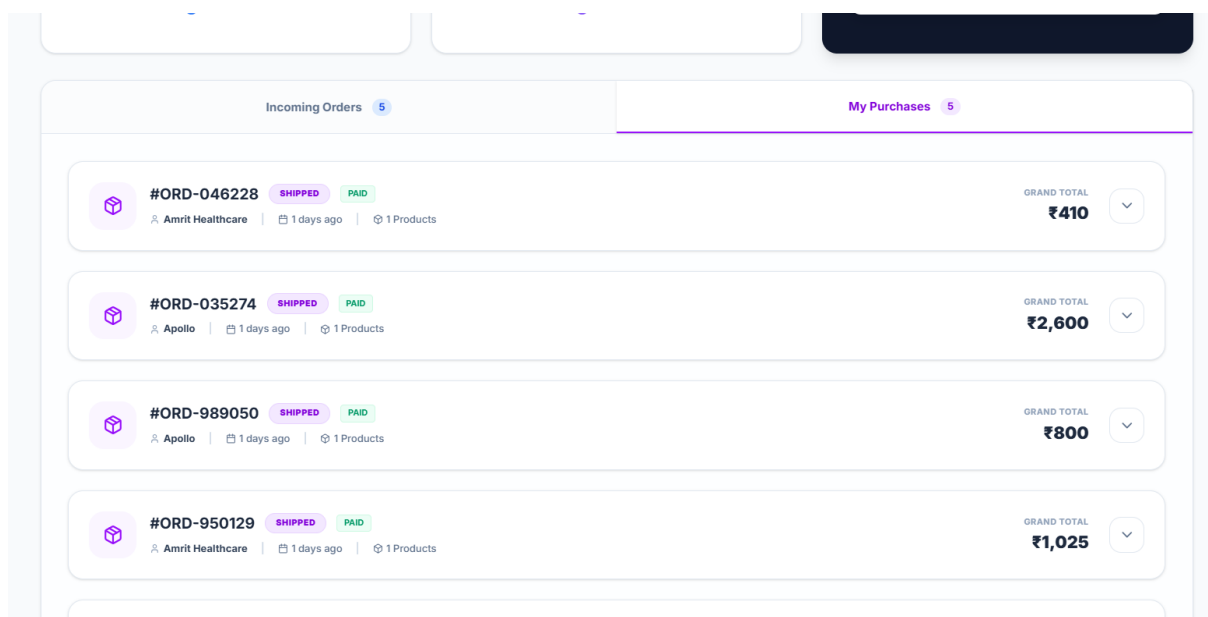


**This figure shows** the detailed view of a distributor order in the MedTrace system, displaying the order status (pending and unpaid), medicine details, quantity, price, billing summary with GST, contact information of the buyer, and options to **approve** or **reject** the order.





**This figure shows** the completed order status in the MedTrace system, where the order has been **confirmed, paid, shipped, and delivered**, along with medicine details, buyer contact information, billing summary, and a message indicating that the order has been successfully completed.



**This figure shows** the *My Purchases* section in the MedTrace system, where the distributor can view a list of previously placed and completed orders from different manufacturers, along with their order status, payment status, order date, and grand total amount.

Incoming Orders 5
My Purchases 5

#ORD-046228

SHIPPED PAID

Amrit Healthcare

1 days ago

1 Products

GRAND TOTAL

₹410

ORDER PLACED

CONFIRMED

SHIPPED & DELIVERED

| MEDICINE | STRENGTH | QTY | PRICE |
|----------|----------|-----|-------|
| Calpol   | 650      | 5   | ₹82   |

CONTACT DETAILS

Amrit Healthcare

+91 98765 43210

contact@medtrace.com

BILLING SUMMARY

Subtotal

₹336.20

Tax (GST 18%)

₹73.80

Grand Total

₹410

Order Completed Successfully

All details are shown here.

MedTrace

DISTRIBUTOR

Authorized Distributor

Dashboard

Track Medicine

LOGISTICS

Incoming Shipments

My Inventory

Manage Orders

Dispatch Stock

Quality & Recall

ROLE

DISTRIBUTOR

Sign Out

SAHA

VERIFIED MANUFACTURER

ADDRESS NOT UPDATED

EDIT PROFILE

DRUG LICENSE

GJ-8647

GST NUMBER

NOT SET

CONTACT

NOT SET

EMAIL

bose04@gmail.com

New Dispatch & Invoice

Create a shipment, generate invoice, and dispatch stock.

INVOICE NUMBER

INV-2026007

SELECT RETAILER

-- Choose Retailer --

DISPATCH DATE

01/31/2026

Add Products

Product

Qty

Price (₹)

Select Medicine

0

0

+ Add

| PRODUCT        | QTY | PRICE | TOTAL | ACTION |
|----------------|-----|-------|-------|--------|
| Cart is empty. |     |       |       |        |

Summary

Subtotal

₹0.00

GST (18%)

₹0.00

Discount

- ₹0.00

Total Payable

₹0.00

Preview Invoice

Confirm & Dispatch

This page is the **Dispatch Stock** interface of the Distributor portal. From here, the distributor can select a retailer, choose the dispatch date, add medicines with quantity and price, and automatically generate an invoice. The system calculates the subtotal, GST, discounts, and total payable amount in real time.

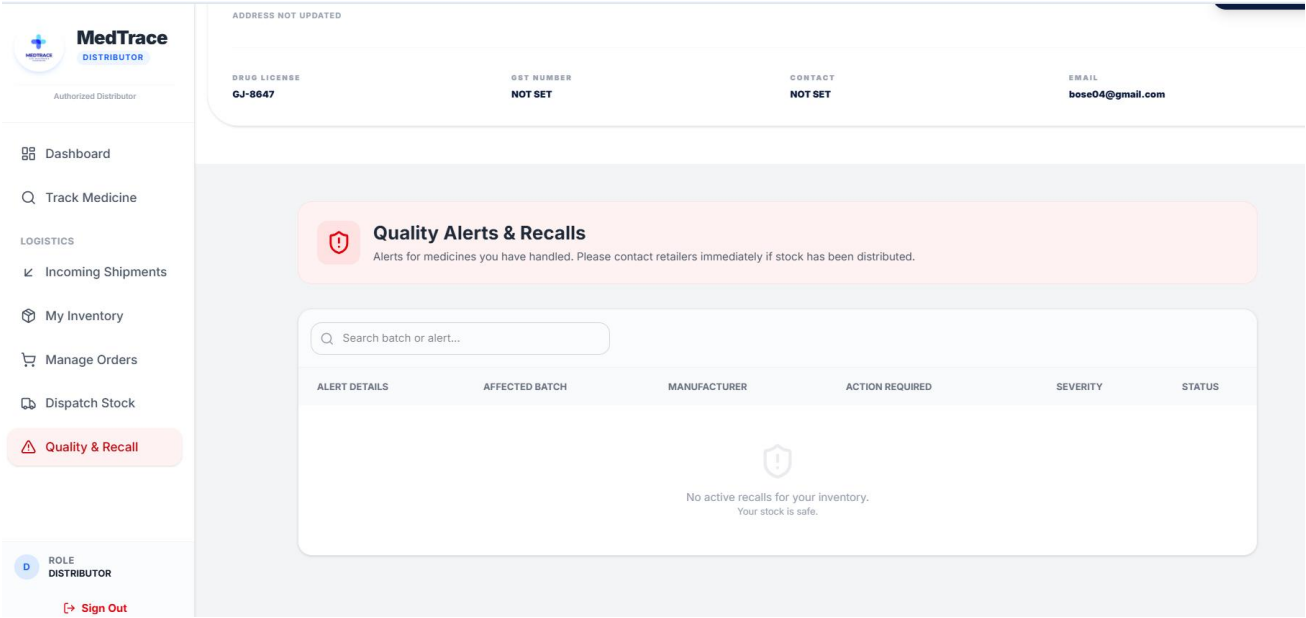
Once all details are entered, the distributor can preview the invoice and confirm the dispatch. After confirmation, the stock is marked as dispatched in the system and the retailer is notified, ensuring proper tracking and transparent movement of medicines through the supply chain.

The screenshot shows the MedTrace distributor interface. On the left is a sidebar with navigation options: Dashboard, Track Medicine, Incoming Shipments, My Inventory, Manage Orders, Dispatch Stock (highlighted), and Quality & Recall. The main area is titled 'New Dispatch & Invoice' with the subtitle 'Create a shipment, generate invoice, and dispatch stock.' The interface includes a header for the distributor 'SAHA' (Verified Manufacturer) with fields for Drug License (GJ-8647), GST Number (Not Set), Contact (Not Set), and Email (bose04@gmail.com). The 'New Dispatch & Invoice' section has a 'SELECT RETAILER' dropdown set to 'Sovan' and a 'DISPATCH DATE' of '01/31/2026'. Below this is an 'Add Products' section with a 'Product' dropdown set to 'Dolo', 'Qty' of '5', and 'Price (₹)' of '₹92'. A table below shows the product details: Dolo (B-202601-004-DICS) with Qty 5, Price ₹92, and Total ₹460. To the right is a 'Summary' section showing Subtotal ₹460.00, GST (18%) ₹82.80, Discount - ₹0.00, and Total Payable ₹542.80. At the bottom right are buttons for 'Preview Invoice' and 'Confirm & Dispatch'.

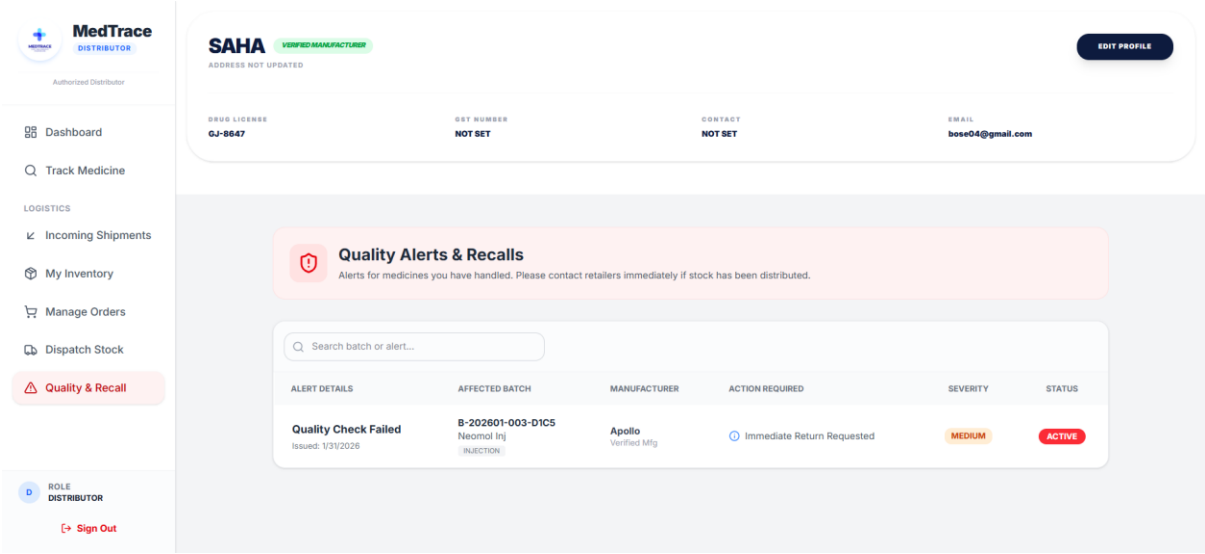
**This current shipment is being created for the retailer Sovan** using the MedTrace system. The distributor selects Sovan, adds the medicine **Dolo** with the required quantity and price, and the system automatically calculates the subtotal, GST, and total payable amount. After reviewing the invoice summary, the distributor confirms the dispatch to complete the shipment.

This screenshot shows the same 'New Dispatch & Invoice' interface as the previous one, but with a confirmation message overlay. The message box from 'my-medicine-app.vercel.app' says 'Shipment Confirmed Successfully!' with an 'OK' button. The background interface remains the same, showing the retailer 'Sovan', product 'Dolo', and the total payable amount of ₹542.80.

**This figure shows** a confirmation message indicating that the shipment to **Sovan** has been **successfully dispatched** through the MedTrace system.

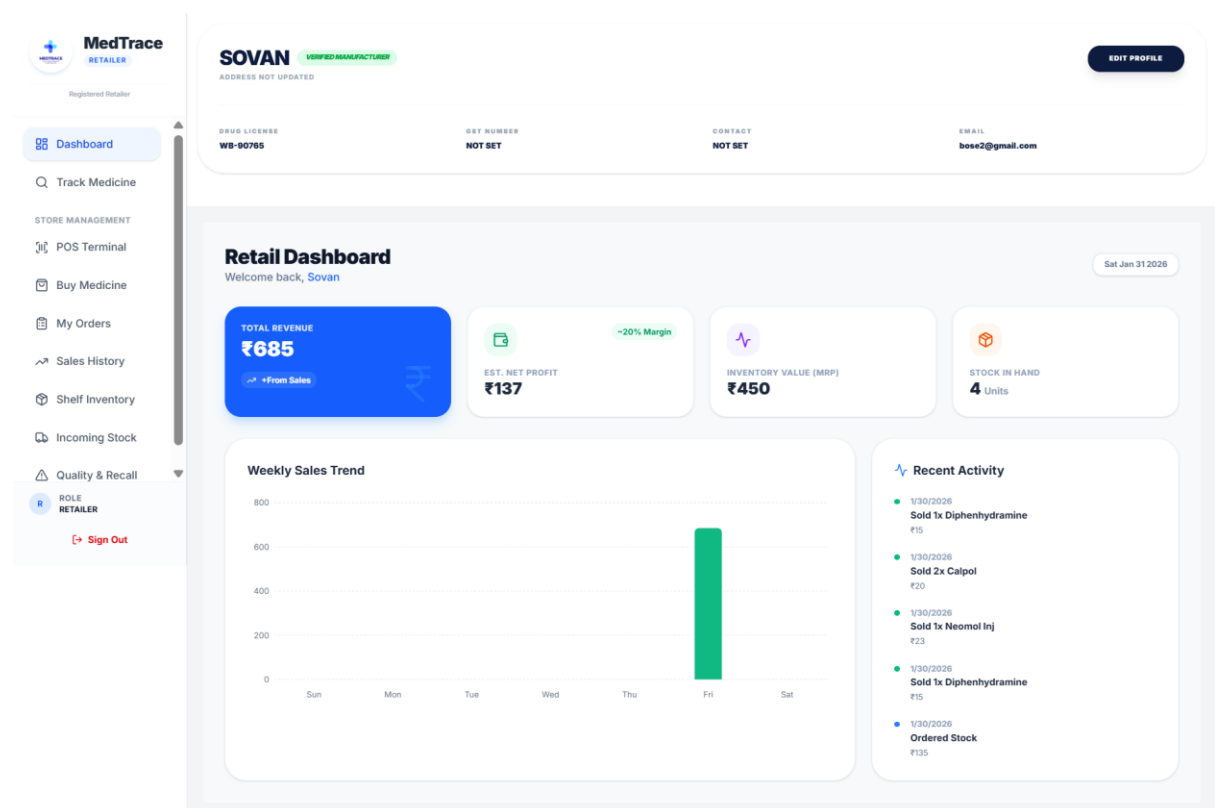


This page is the **Quality Alerts & Recalls** section of the Distributor portal. If any medicine batch is recalled by the manufacturer, all related details are displayed here in real time. The distributor can view the affected batch number, manufacturer name, severity level, and the action required. This screen ensures that distributors are immediately informed about safety issues. They can quickly stop further distribution, isolate the recalled stock, and contact retailers who may have already received the affected medicines, helping to prevent unsafe products from reaching consumers.



**This figure shows** the *Quality Alerts & Recalls* section in the MedTrace distributor panel, where a recalled medicine batch (**Neomol Inj**, Batch **B-202601-003-D1C5**) is listed as unsafe due to a **quality check failure**. The alert indicates the manufacturer (Apollo), the required action (**immediate return requested**), the severity level, and the active recall status, allowing the distributor to take quick corrective action.

### 3. Retailer Login Page

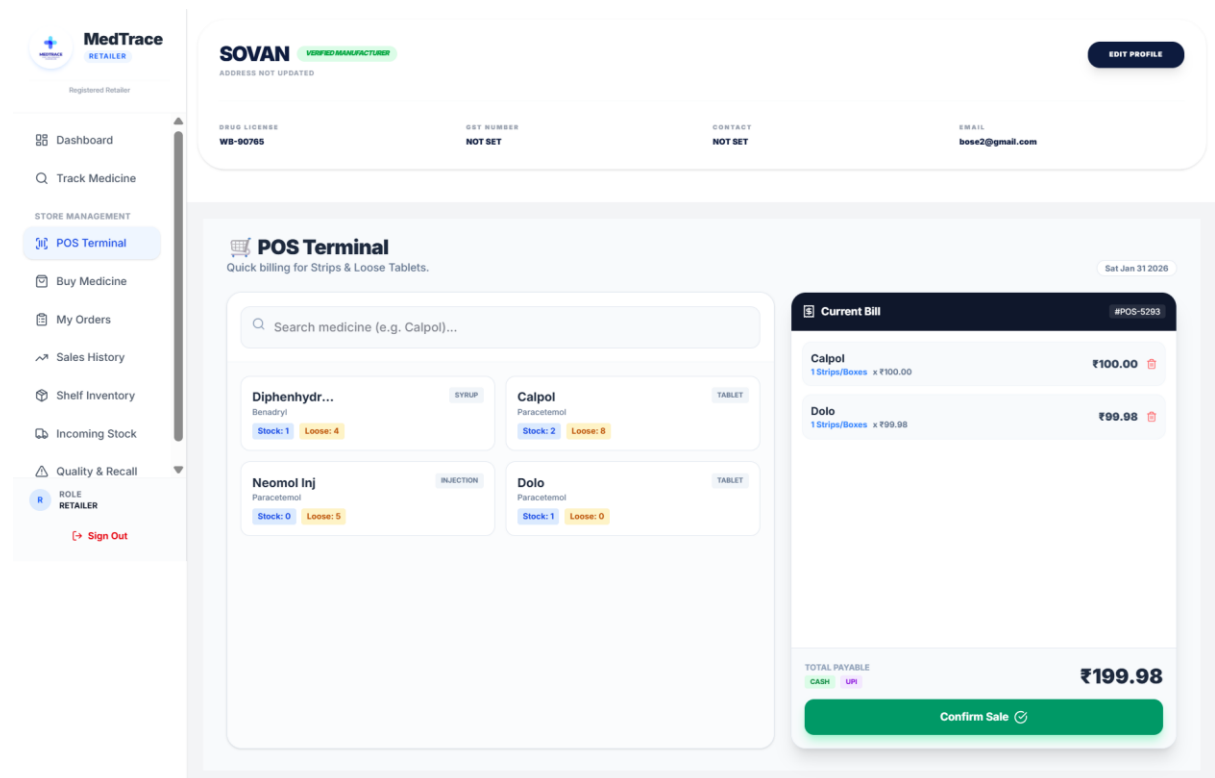


This is the **Retailer Dashboard** of the MedTrace system. After logging in, the retailer can see a complete overview of their shop's activity. The dashboard shows total sales, estimated profit, inventory value, and current stock in hand, giving a quick summary of business performance.

The **Weekly Sales Trend** graph helps the retailer understand daily sales patterns, while the **Recent Activity** panel displays the latest transactions such as medicines sold and stock ordered. From the left-side menu, the retailer can access features like POS billing, buying medicine from distributors, managing orders, tracking shelf inventory, checking incoming stock, and viewing quality or recall alerts.



This dashboard helps retailers manage stock efficiently, track sales, and immediately respond to recalled or unsafe medicines.

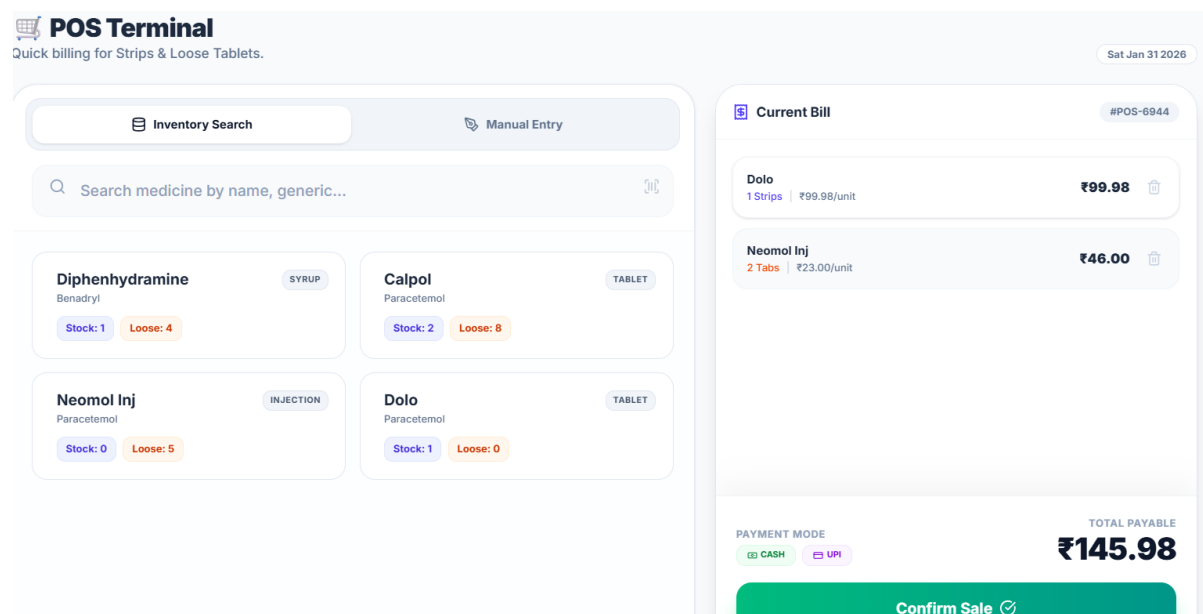


This page is the **POS (Point of Sale) Terminal** for the retailer. It is used for quick and easy billing of medicines, whether they are sold as full strips/boxes or as loose tablets.

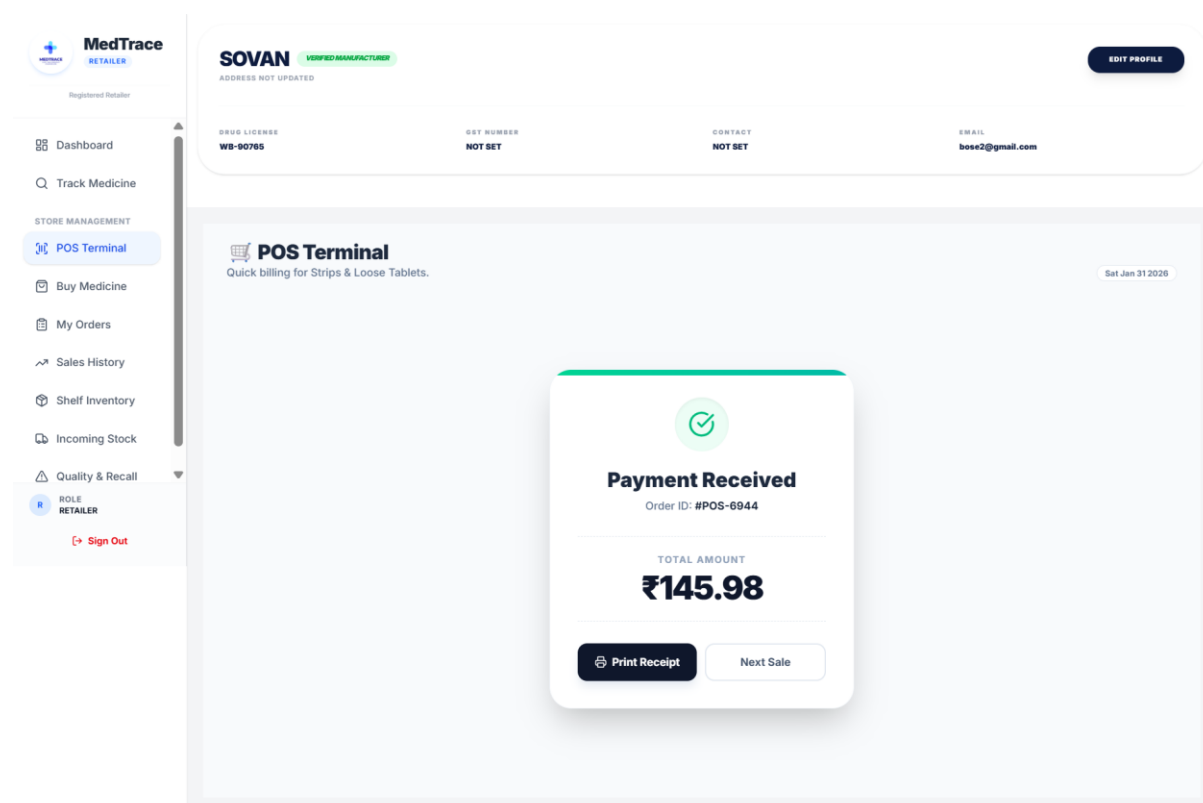
On the left side, the retailer can **search and select medicines** such as Calpol, Dolo, or Diphenhydramine. Each medicine card shows the available **stock and loose quantity**, so the seller knows exactly how much is left before adding items to the bill.

On the right side, the **Current Bill** panel displays all selected medicines, their quantities, individual prices, and the total amount. The system automatically calculates the **total payable amount** and allows the retailer to choose a payment mode such as **Cash or UPI**.

Once the sale is confirmed by clicking **“Confirm Sale”**, the stock is updated in real time, the transaction is recorded, and the sale appears in the retailer’s sales history.



**This figure shows** the retailer POS terminal section of the MedTrace system, where the selling of medicines takes place by selecting products from inventory, adding them to the current bill, choosing the payment mode, and confirming the sale.



**This figure shows** the payment confirmation screen of the retailer POS terminal in the MedTrace system, indicating that the sale has been successfully completed, displaying the order ID, total amount paid, and options to print the receipt or proceed to the next sale.

MedTrace

REGISTERED RETAILER

Dashboard

Track Medicine

STORE MANAGEMENT

POS Terminal

Buy Medicine

My Orders

Sales History

Shelf Inventory

Incoming Stock

Quality & Recall

ROLE

RETAILER

Sign Out

SOVAN

VERIFIED MANUFACTURER

ADDRESS NOT UPDATED

EDIT PROFILE

PRD LICENSED

WB-90765

GST NUMBER

NOT SET

CONTACT

NOT SET

EMAIL

bose2@gmail.com

Medicine Marketplace

Verified B2B Procurement Portal

LIVE INVENTORY

7 Products

Q

Search by brand name, generic composition...

Lowest Price

Highest Price

D

Diphenhydramine

BENADRYL

QUALITY VERIFIED

SYRUP

MANUFACTURER

Apollo

BEST MARKET PRICE

₹135.00

SAVE 5%

AVAILABILITY

2 Distributors

Prices vary by distributor. Compare before buying.

Compare 2 Sellers

N

Neomol Inj

PARACETEMOL

QUALITY VERIFIED

INJECTION

MANUFACTURER

Apollo

BEST MARKET PRICE

₹220.00

SAVE 6%

AVAILABILITY

1 Distributors

Prices vary by distributor. Compare before buying.

Check Stock & Buy

D

Dolo

PARACETEMOL

QUALITY VERIFIED

TABLET

MANUFACTURER

Apollo

BEST MARKET PRICE

₹85.00

SAVE 5%

AVAILABILITY

1 Distributors

Prices vary by distributor. Compare before buying.

Check Stock & Buy

C

Calpol

PARACETEMOL

QUALITY VERIFIED

TABLET

MANUFACTURER

Amrit Healthcare

BEST MARKET PRICE

₹89.00

SAVE 1%

AVAILABILITY

2 Distributors

Prices vary by distributor. Compare before buying.

Compare 2 Sellers

G

Gaspaz

DOMPERIDONE AND OMEPRAZOLE

QUALITY VERIFIED

CAPISULE

MANUFACTURER

bose

BEST MARKET PRICE

₹190.00

SAVE 5%

AVAILABILITY

1 Distributors

Prices vary by distributor. Compare before buying.

Check Stock & Buy

N

Neomol Inj

PARACETEMOL

QUALITY VERIFIED

INJECTION

MANUFACTURER

Amrit Healthcare

BEST MARKET PRICE

₹215.00

SAVE 2%

AVAILABILITY

1 Distributors

Prices vary by distributor. Compare before buying.

Check Stock & Buy

D

DOLO

PARACETAMOL

QUALITY VERIFIED

TABLET

MANUFACTURER

bose

BEST MARKET PRICE

₹90.00

SAVE 5%

AVAILABILITY

1 Distributors

Prices vary by distributor. Compare before buying.

Check Stock & Buy

This page is the **Medicine Marketplace** for retailers, where they can purchase medicines directly from verified distributors at the best available prices.

The retailer can **search medicines by brand name or generic composition** and also use filters such as **Lowest Price** and **Highest Price** to compare options easily. Each product card shows the **manufacturer name, best market price, availability (number of distributors), and quality verification status**.

Retailers can:

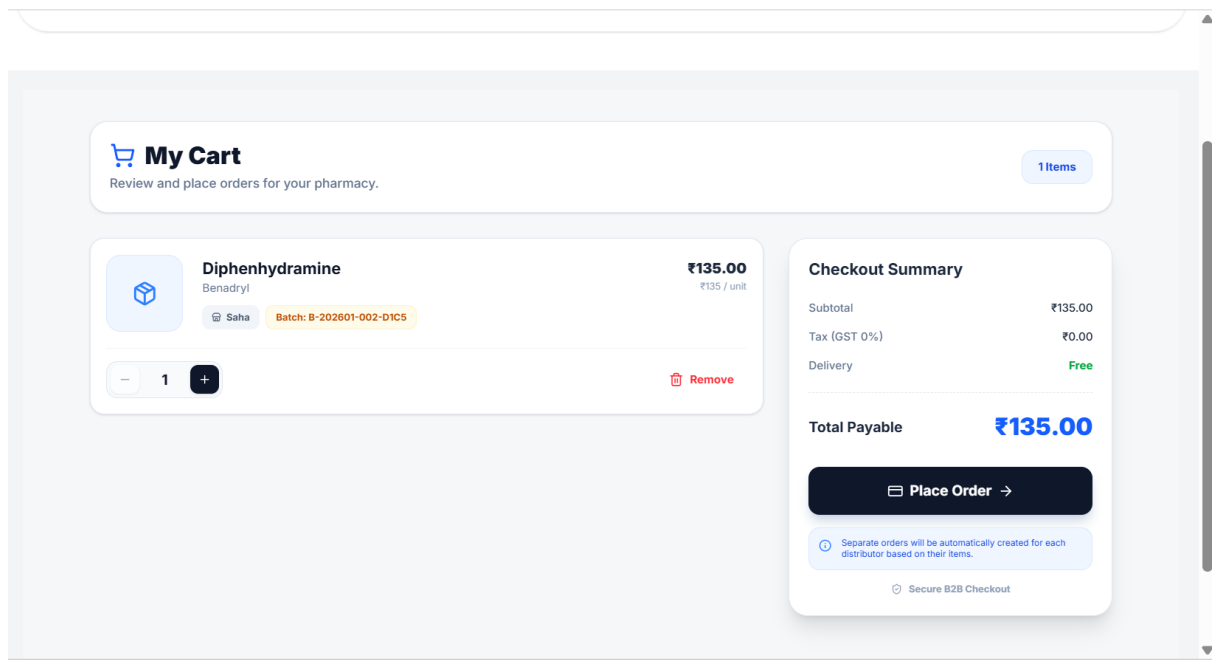
- **Compare sellers** to choose the most cost-effective option.
- **Check stock and buy** directly from a selected distributor.
- View multiple offers for the same medicine from different suppliers.

This marketplace ensures **transparent pricing, verified quality, and trusted sourcing**, helping retailers restock quickly while reducing the risk of counterfeit or expired medicines.

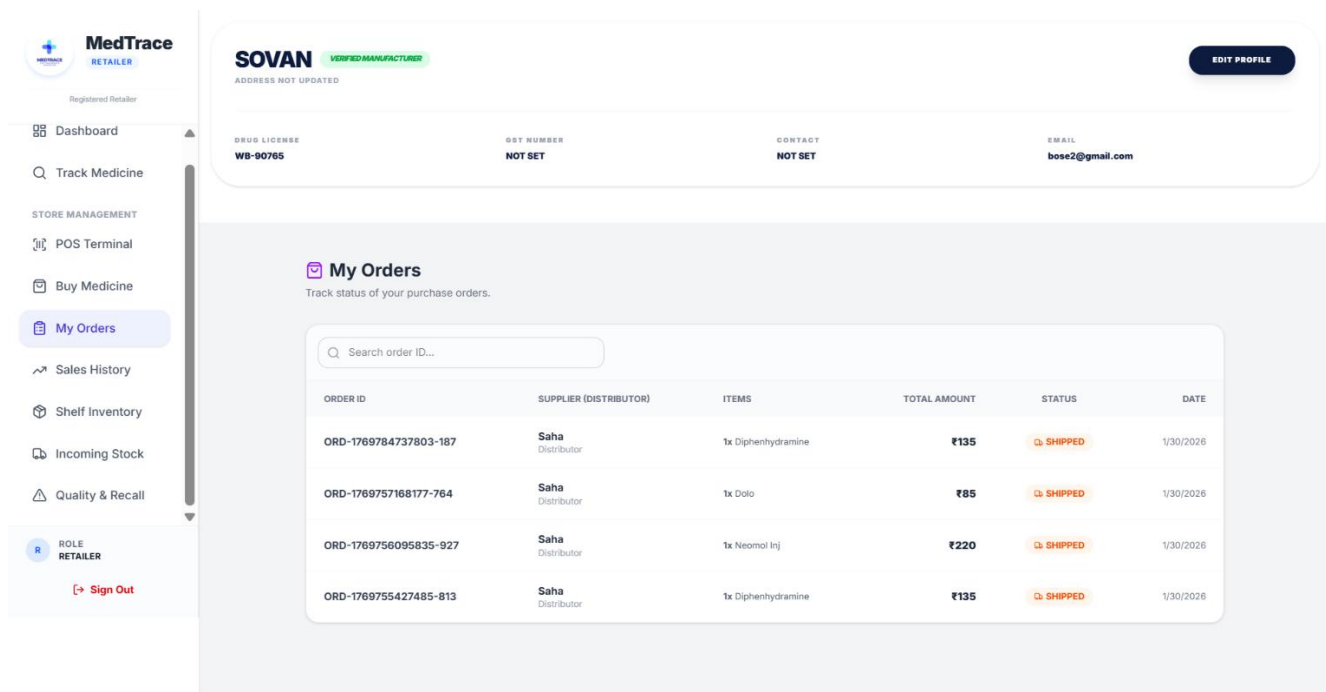
The screenshot displays the MedTrace Retailer interface. On the left is a sidebar with navigation links: Dashboard, Track Medicine, STORE MANAGEMENT (POS Terminal), Buy Medicine, My Orders, Sales History, Shelf Inventory, Incoming Stock, and Quality & Recall. The main content area shows the product page for Diphenhydramine Syrup (Benadryl • 150 mg). At the top, it indicates 'AVAILABLE FROM 2 Distributors'. Below this, there's a 'Select a Distributor' section with two options: Saha and Roy. Each distributor card shows their profile picture, name, location (N/A), verification status (Verified), available stock (18 units for Saha, 20 units for Roy), expiry date (Jan 2026), and the best market price (₹135.00 for Saha, ₹140.00 for Roy). Both prices are shown with a discount percentage (90% OFF for Saha, 7% OFF for Roy) and a 'Buy Now' button. The top header includes the MedTrace logo, 'REGISTERED RETAILER', and an 'EDIT PROFILE' button. Below the header, there's a section for 'DRUG LICENSE', 'GST NUMBER', 'CONTACT', and 'EMAIL'.

| Distributor | Available Stock | Expiry Date | Best Market Price | Discount |
|-------------|-----------------|-------------|-------------------|----------|
| Saha        | 18 Units        | Jan 2026    | ₹135.00           | 90% OFF  |
| Roy         | 20 Units        | Jan 2026    | ₹140.00           | 7% OFF   |

**This figure shows** that the same medicine (*Diphenhydramine syrup*) is available from **two different distributors**. Although the product and strength are the same, each distributor offers it at a **different price**. The MedTrace system allows the retailer to compare stock, expiry date, and price, and then choose the **lowest-priced option** to reduce cost while restocking.



**This figure shows** the *My Cart* page in the MedTrace retailer panel, where the selected medicine (**Diphenhydramine**) from distributor **Saha** is added to the cart with its batch number and price. The retailer can adjust the quantity, review the checkout summary, and place the order by clicking **Place Order**.



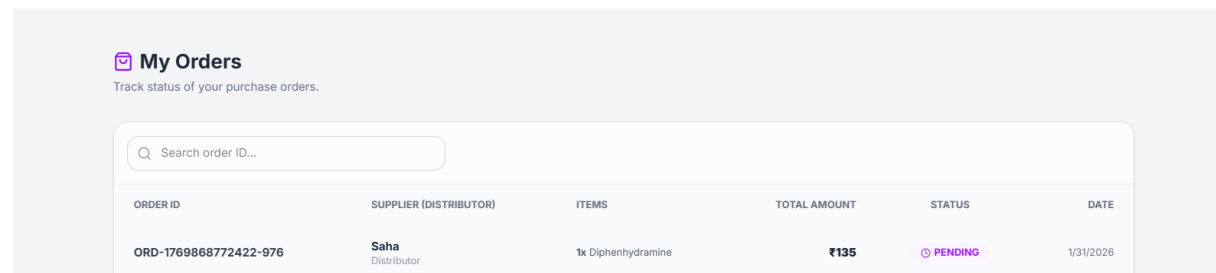
This page allows the retailer to **track all purchase orders** made from the marketplace.

Each order shows the **order ID, distributor name, medicine name, total amount, status (shipped/pending), and order date.**

From this section, the retailer can:

- View the **current delivery status** of each order
- Check which **distributor supplied the medicine**
- Track **how much was paid** and **what items were ordered**

This helps the retailer monitor their supply flow and ensure timely stock replenishment.



**My Orders**  
Track status of your purchase orders.

Q Search order ID...

| ORDER ID              | SUPPLIER (DISTRIBUTOR)     | ITEMS              | TOTAL AMOUNT | STATUS                                                                                      | DATE      |
|-----------------------|----------------------------|--------------------|--------------|---------------------------------------------------------------------------------------------|-----------|
| ORD-1769868772422-976 | <b>Saha</b><br>Distributor | 1x Diphenhydramine | ₹135         |  PENDING | 1/31/2026 |

**This figure shows** the *My Orders* page in the MedTrace retailer panel, where the order placed to distributor **Saha** for **Diphenhydramine** is displayed. After the distributor approved the order in the distributor section, the order status is automatically updated and reflected here, allowing the retailer to track the current progress of the purchase.

**MedTrace**  
REGISTERED RETAILER

Registered Retailer

Dashboard

Track Medicine

STORE MANAGEMENT

POS Terminal

Buy Medicine

My Orders

Sales History

Shelf Inventory

Incoming Stock

Quality & Recall

ROLE  
RETAILER

Sign Out

**SOVAN**  
VERIFIED MANUFACTURER

ADDRESS NOT UPDATED

EDIT PROFILE

DRUG LICENSE  
**WB-90765**

GST NUMBER  
**NOT SET**

CONTACT  
**NOT SET**

EMAIL  
**bose2@gmail.com**

**Sales History**  
Track every transaction and monitor your revenue.

TOTAL REVENUE  
**₹ 685**

Search by medicine name, batch no...

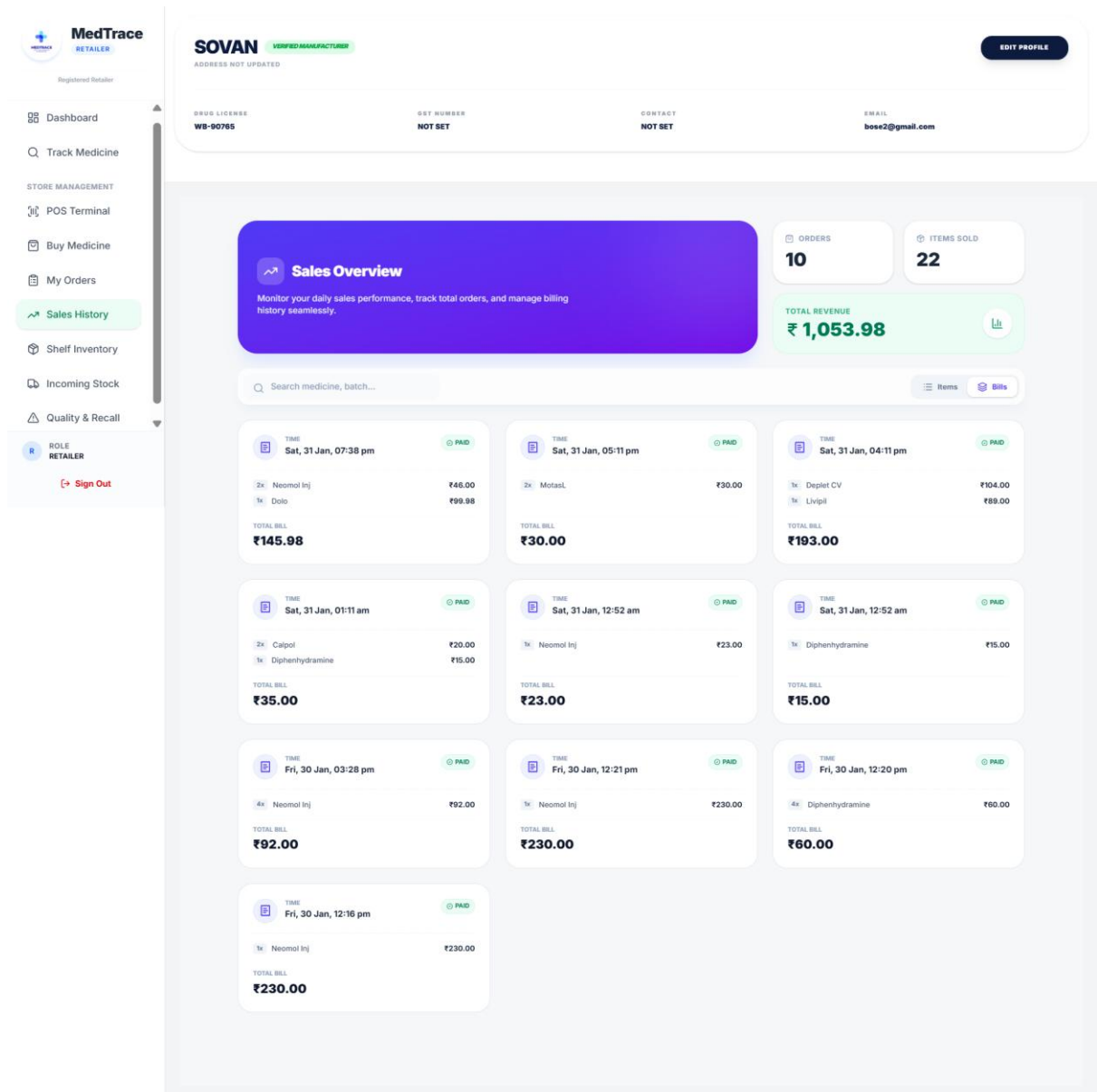
Filter

| DATE & TIME           | MEDICINE INFO                                 | BATCH NO          | QUANTITY | AMOUNT                        |
|-----------------------|-----------------------------------------------|-------------------|----------|-------------------------------|
| 1/30/2026<br>07:41 PM | <b>Diphenhydramine</b><br>Benadryl<br>SYRUP   | B-202601-002-D1C5 | 1 TABS   | ₹ 15.00<br>Paid via Cash/UPI  |
| 1/30/2026<br>07:41 PM | <b>Calpol</b><br>Paracetamol<br>TABLET        | B-202601-002-6AE2 | 2 TABS   | ₹ 20.00<br>Paid via Cash/UPI  |
| 1/30/2026<br>07:22 PM | <b>Neomol Inj</b><br>Paracetamol<br>INJECTION | B-202601-003-D1C5 | 1 TABS   | ₹ 23.00<br>Paid via Cash/UPI  |
| 1/30/2026<br>07:22 PM | <b>Diphenhydramine</b><br>Benadryl<br>SYRUP   | B-202601-002-D1C5 | 1 TABS   | ₹ 15.00<br>Paid via Cash/UPI  |
| 1/30/2026<br>09:58 AM | <b>Neomol Inj</b><br>Paracetamol<br>INJECTION | B-202601-003-D1C5 | 4 TABS   | ₹ 92.00<br>Paid via Cash/UPI  |
| 1/30/2026<br>06:51 AM | <b>Neomol Inj</b><br>Paracetamol<br>INJECTION | B-202601-003-D1C5 | 1 STRIPS | ₹ 230.00<br>Paid via Cash/UPI |
| 1/30/2026<br>06:50 AM | <b>Diphenhydramine</b><br>Benadryl<br>SYRUP   | B-202601-002-D1C5 | 4 STRIPS | ₹ 60.00<br>Paid via Cash/UPI  |
| 1/30/2026<br>06:46 AM | <b>Neomol Inj</b><br>Paracetamol<br>INJECTION | B-202601-003-D1C5 | 1 STRIPS | ₹ 230.00<br>Paid via Cash/UPI |

This page displays the **complete sales record** of the retailer in real time. Each entry shows the **date & time, medicine name, batch number, quantity sold, and total amount received**.

It also shows the **total revenue** at the top, helping the retailer track daily earnings.

Using the **search and filter options**, the retailer can quickly find any past transaction.



**This figure shows** the *Sales History* dashboard in the MedTrace retailer panel, where the retailer can view all completed sales with time, product details, total bill amount, and payment status. It also displays a sales overview summary including total orders, items sold, and total revenue, helping the retailer track daily sales performance and manage store billing records efficiently.



**MedTrace**  
REGISTERED RETAILER

Registered Retailer

Dashboard

Track Medicine

STORE MANAGEMENT

POS Terminal

Buy Medicine

My Orders

Sales History

**Shelf Inventory**

Incoming Stock

Quality & Recall

ROLE  
RETAILER

Sign Out

**SOVAN**
VERIFIED MANUFACTURER

ADDRESS NOT UPDATED

EDIT PROFILE

DRUG LICENSE  
**WB-90765**

GST NUMBER  
**NOT SET**

CONTACT  
**NOT SET**

EMAIL  
**bese2@gmail.com**

**My Shelf Inventory**

Total Items: 4

**Diphenhydramine**
SYRUP

Benadryl

Batch: **B-202601-002-D1C5**  
Expiry: **1/20/2026**

CURRENT STOCK  

1 Strips  
+ 4 Loose Tabs

Low Stock Alert

**Calpol**
TABLET

Paracetamol

Batch: **B-202601-002-6AE2**  
Expiry: **1/1/2027**

CURRENT STOCK  

2 Strips  
+ 8 Loose Tabs

Low Stock Alert

**Neomol Inj**
INJECTION

Paracetamol

Batch: **B-202601-003-D1C5**  
Expiry: **2/1/2027**

CURRENT STOCK  

0 Strips  
+ 5 Loose Tabs

Low Stock Alert

**Dolo**
TABLET

Paracetamol

Batch: **B-202601-004-D1C5**  
Expiry: **2/2/2026**

CURRENT STOCK  

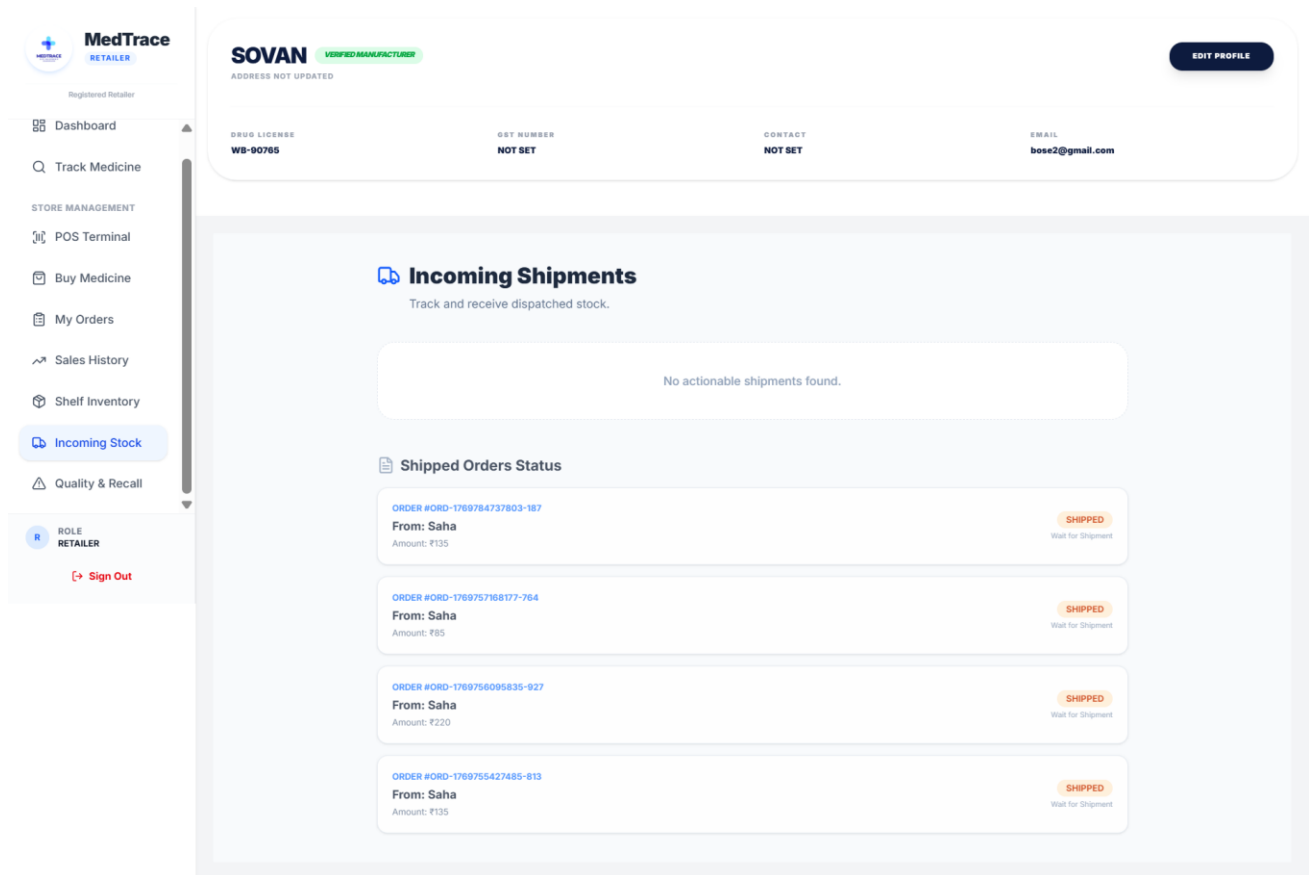
1 Strips

Low Stock Alert

This page shows the **live shelf stock** available in the retail store. Each card displays the **medicine name, batch number, expiry date, and current stock** (strips and loose tablets).

The system automatically highlights **low stock alerts** and also shows **expiry information**, helping the retailer avoid selling expired or near-expiry medicines.

This feature allows the retailer to **monitor shelf availability in real time**, plan re-orders on time, and ensure safe medicine dispensing to customers.

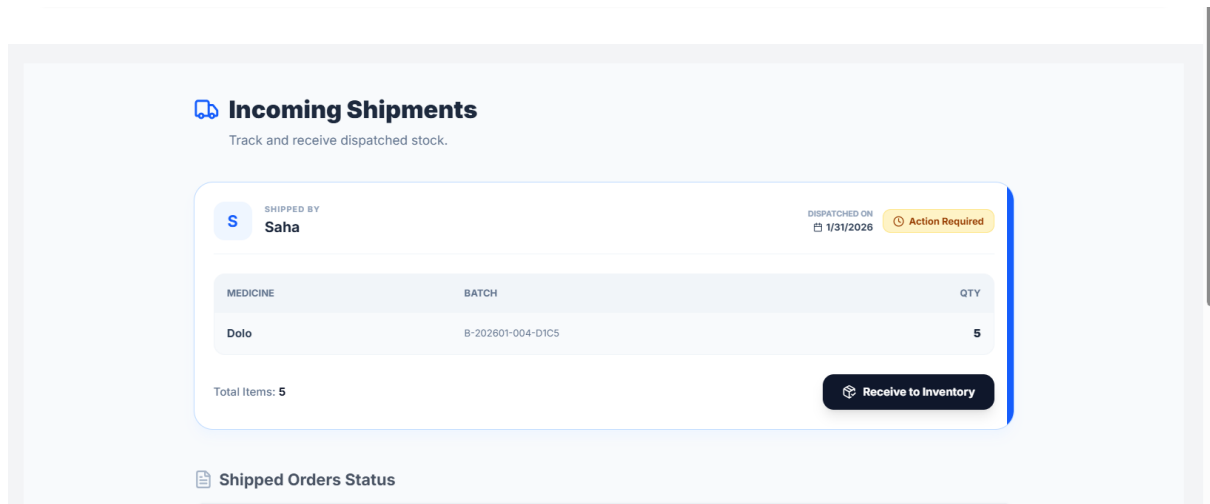


This page allows the retailer to **track all incoming shipments** that have been dispatched by the distributor.

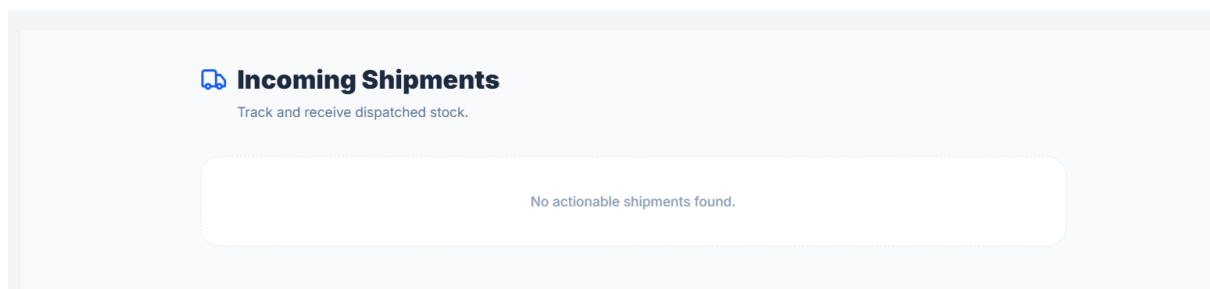
Each entry shows the **order ID, supplier name, order amount, and shipment status**.

When the status is marked as **Shipped**, it means the stock is on the way and the retailer is waiting to receive it. Once the shipment is delivered, it can be **added to shelf inventory** automatically.

This feature helps the retailer **monitor pending deliveries**, avoid stock shortages, and maintain a smooth medicine supply.



**This figure shows** the *Incoming Shipments* page in the MedTrace retailer panel, where a dispatched order from distributor **Saha** is listed. It displays the medicine name (**Dolo**), batch number, quantity, dispatch date, and an action button to **receive the stock into inventory**.



**After accepting the shipment,** the *Incoming Shipments* page updates to show **“No actionable shipments found,”** indicating that the stock has been successfully received and added to the retailer’s inventory.

The screenshot displays the MedTrace system interface for a registered retailer. The sidebar on the left contains navigation links: Dashboard, Track Medicine, POS Terminal, Buy Medicine, My Orders, Sales History, Shelf Inventory, Incoming Stock, and Quality & Recall (highlighted). The main content area shows the retailer's profile for 'SOVAN' (Verified Manufacturer) with fields for Drug License (WB-90765), Set Number (NOT SET), Contact (NOT SET), and Email (bosco2@gmail.com). Below this, a 'Critical Medicine Alerts' section is visible, featuring a search bar and a table with columns: REASON FOR RECALL, MEDICINE & BATCH, ACTION REQUIRED, SEVERITY, and STATUS. The table is currently empty, displaying a message: 'No active recalls found. Your inventory is currently safe to sell.'

This page displays **critical medicine alerts** sent by manufacturers through the MedTrace system.

If any batch is recalled, the retailer receives an **immediate warning** with full details such as the **medicine name, batch number, reason for recall, severity, and required action**.

The system instructs the retailer to **stop selling the affected medicine instantly**, remove it from shelves, and return or isolate the stock. When there are no recalls, the page clearly shows that the **inventory is safe to sell**, giving confidence to the retailer and protecting patient safety.

MedTrace

RETAILER

Registered Retailer

Dashboard

Track Medicine

STORE MANAGEMENT

POS Terminal

Buy Medicine

My Orders

Sales History

Shelf Inventory

Incoming Stock

Quality & Recall

ROLE  
RETAILER

Sign Out

SOVAN

VERIFIED MANUFACTURER

ADDRESS NOT UPDATED

EDIT PROFILE

DRUG LICENSE  
WB-90765

GST NUMBER  
NOT SET

CONTACT  
NOT SET

EMAIL  
bose2@gmail.com

Critical Medicine Alerts

Stop selling these items immediately. Remove affected batches from shelves and return them.

Search medicine name or batch...

| REASON FOR RECALL                                          | MEDICINE & BATCH                                                                 | ACTION REQUIRED                                          | SEVERITY               | STATUS            |
|------------------------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------|------------------------|-------------------|
| <div>Quality Check Failed</div> <div>Date: 1/31/2026</div> | <div>Neomol Inj</div> <div>Batch: B-202601-003-D1C5</div> <div>Mfg: Apollo</div> | <div>Remove from shelf &amp; Return to Distributor</div> | <div>MEDIUM RISK</div> | <div>Active</div> |

**This figure shows** the *Critical Medicine Alerts* section in the MedTrace retailer panel, where a recalled medicine batch (**Neomol Inj**, Batch **B-202601-003-D1C5**) is flagged due to a **quality check failure**. The alert instructs the retailer to **stop selling, remove the batch from shelves, and return it to the distributor**, ensuring patient safety and regulatory compliance.

## **IV. Conclusion**

The MedTrace system provides a complete, end-to-end digital solution for medicine tracking and verification across the entire supply chain. From the moment a medicine is manufactured and assigned a unique QR code, to the time it reaches the consumer and is scanned for authenticity, every step is recorded, monitored, and verified in real time.

By connecting manufacturers, distributors, retailers, and consumers on a single platform, MedTrace ensures full transparency and traceability. The system not only helps in identifying genuine medicines, but also plays a critical role in detecting expired, recalled, or fake products before they can cause harm.

One of the most powerful features of this system is the instant recall mechanism, where defective medicines can be flagged immediately and all connected stakeholders receive alerts. This allows rapid removal of unsafe products from the market, protecting public health and reducing risk.

Overall, MedTrace strengthens trust in the pharmaceutical supply chain, improves safety, reduces counterfeit circulation, and enables faster, smarter decision-making. It represents a major step toward a secure, transparent, and digitally connected healthcare ecosystem.

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